

MATHEMATICS BASIC 2024 ALL SETS

Mathematics (Basic) - Chapter-wise MCQ Questions Bank

Chapter 1: Real Numbers

YEAR: 2024

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the HCF of two positive integers a and b is 1, then their LCM is:

OPTIONS: (A) $a+b$ (B) a (C) b (D) ab

ANSWER: (D) ab

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number $3 + \sqrt{2}$ is:

OPTIONS: (A) a rational number (B) an irrational number (C) an integer (D) a natural number

ANSWER: (B) an irrational number

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: HCF (132, 77) is:

OPTIONS: (A) 11 (B) 77 (C) 22 (D) 44

ANSWER: (A) 11

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $1080 = 2^x \times 3^y \times 5$, then $(x - y)$ is equal to:

OPTIONS: (A) 6 (B) -1 (C) 1 (D) 0

ANSWER: (C) 1

YEAR: 2024

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number 3^n , where n is a natural number, cannot end with the digit:

OPTIONS: (A) 1 (B) 3 (C) 5 (D) 7

ANSWER: (C) 5

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A prime number has:

OPTIONS: (A) exactly two prime factors (B) exactly one prime factor (C) at least one prime factor (D) at least two prime factors

ANSWER: (B) exactly one prime factor

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $p = 2^3 \times 3^2 \times 5$ and $q = 2^2 \times 3^3$, then the LCM of p and q is:

OPTIONS: (A) $2^3 \times 3^3$ (B) $2^2 \times 3^2$ (C) $2^2 \times 3^2 \times 5$ (D) $2^3 \times 3^3 \times 5$

ANSWER: (D) $2^3 \times 3^3 \times 5$

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of (HCF - LCM) for the two numbers 3 and 5 is:

OPTIONS: (A) 2 (B) 4 (C) 14 (D) -14

ANSWER: (D) -14

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number 2^n , where n is a natural number, cannot end with the digit:

OPTIONS: (A) 4 (B) 6 (C) 2 (D) 0

ANSWER: (D) 0

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $(2 + \sqrt{2})^2$ is:

OPTIONS: (A) a rational number (B) an irrational number (C) an integer (D) a natural number

ANSWER: (B) an irrational number

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: HCF (850, 500) is:

OPTIONS: (A) 850×50 (B) 17×500 (C) $17 \times 5^2 \times 2^2$ (D) $17 \times 5^3 \times 2$

ANSWER: (C) $17 \times 5^2 \times 2^2$

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The HCF of smallest 2-digit number and the smallest composite number is:

OPTIONS: (A) 2 (B) 20 (C) 40 (D) 4

ANSWER: (D) 4

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\text{HCF}(x, 20) = 2$ and $\text{LCM}(x, 20) = 60$, then value of x is:

OPTIONS: (A) 3 (B) 6 (C) 20 (D) 10

ANSWER: (B) 6

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The LCM of two numbers is 3600. Which of the following cannot be their HCF?

OPTIONS: (A) 600 (B) 400 (C) 500 (D) 150

ANSWER: (C) 500

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If n is any natural number, then which of the following numbers ends with digit 0?

OPTIONS: (A) $(3 \times 2)^n$ (B) $(5 \times 2)^n$ (C) $(6 \times 2)^n$ (D) $(4 \times 2)^n$

ANSWER: (B) $(5 \times 2)^n$

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Prime factorisation of 424 is:

OPTIONS: (A) $2 \times 53 \times 4$ (B) $2 \times 53 \times 2$ (C) $2^3 \times 53$ (D) $2^4 \times 53$

ANSWER: (C) $2^3 \times 53$

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If x is a whole number, then 8^x ends with an even digit, except for which value of x ?

OPTIONS: (A) 6 (B) 4 (C) 2 (D) 0

ANSWER: (D) 0

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the number a , where n is a natural number, always ends with digit a , then the possible value of a is:

OPTIONS: (A) 2 (B) 4 (C) 6 (D) 8

ANSWER: (C) 6

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Show that 45^n cannot end with the digit 0, n being a natural number. Write the prime number a which on multiplying with 45^n makes the product end with the digit 0.

ANSWER: $45^n = 3^{2n} \times 5^n$. To end with digit 0, 45^n should have prime factors 2 and 5 both. So it cannot end with digit 0. 45^n should be multiplied by 2 $\Rightarrow a = 2$.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Prove that $6 - 4\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number.

ANSWER: Let $6 - 4\sqrt{5}$ be a rational number. Then $4\sqrt{5} = 6 - \text{rational}$, so $\sqrt{5}$ would be rational, which contradicts given that $\sqrt{5}$ is irrational. Hence $6 - 4\sqrt{5}$ is irrational.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Show that $11 \times 19 \times 23 + 3 \times 11$ is not a prime number.

ANSWER: $11 \times 19 \times 23 + 3 \times 11 = 11(19 \times 23 + 3) = 11(437 + 3) = 11 \times 440 = 11 \times 440$. Since it has factors other than 1 and itself, it is not a prime number.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $\sqrt{3}$ is an irrational number.

ANSWER: Let $\sqrt{3}$ be a rational number such that $\sqrt{3} = p/q$ (p and q are co-prime numbers, $q \neq 0$). $\sqrt{3} q = p \Rightarrow 3q^2 = p^2$. 3 divides $p^2 \Rightarrow 3$ divides p as well. Let $p = 3m$ (for some integer m). $3q^2 = 9m^2 \Rightarrow q^2 = 3m^2$. 3 divides $q^2 \Rightarrow 3$ divides q as well. p and q have a common factor 3, which is a contradiction as p and q are co-prime. Hence $\sqrt{3}$ is an irrational number.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $6 + 3\sqrt{2}$ is an irrational number, given that $\sqrt{2}$ is an irrational number.

ANSWER: Let $6 + 3\sqrt{2}$ be a rational number. Then $3\sqrt{2} = \text{rational} - 6$, so $\sqrt{2}$ would be rational, which contradicts given that $\sqrt{2}$ is irrational. Hence $6 + 3\sqrt{2}$ is irrational.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $7 - 3\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number.

ANSWER: Let $7 - 3\sqrt{5}$ be a rational number. Then $3\sqrt{5} = 7 - \text{rational}$, so $\sqrt{5}$ would be rational, which contradicts given that $\sqrt{5}$ is irrational. Hence $7 - 3\sqrt{5}$ is irrational.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $5\sqrt{2} - 3$ is an irrational number, given that $\sqrt{2}$ is an irrational number.

ANSWER: Let $5\sqrt{2} - 3$ be a rational number. Then $5\sqrt{2} = \text{rational} + 3$, so $\sqrt{2}$ would be rational, which contradicts given that $\sqrt{2}$ is irrational. Hence $5\sqrt{2} - 3$ is irrational.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $2 + 5\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.

ANSWER: Let $2 + 5\sqrt{3}$ be a rational number. Then $5\sqrt{3} = \text{rational} - 2$, so $\sqrt{3}$ would be rational, which contradicts given that $\sqrt{3}$ is irrational. Hence $2 + 5\sqrt{3}$ is irrational.

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CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $6 - 4\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number.

ANSWER: Let $6 - 4\sqrt{5}$ be a rational number. Then $4\sqrt{5} = 6 - \text{rational}$, so $\sqrt{5}$ would be rational, which contradicts given that $\sqrt{5}$ is irrational. Hence $6 - 4\sqrt{5}$ is irrational.

YEAR: 2024

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $3 - 2\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number.

ANSWER: Let $3 - 2\sqrt{5}$ be a rational number. Then $2\sqrt{5} = 3 - \text{rational}$, so $\sqrt{5}$ would be rational, which contradicts given that $\sqrt{5}$ is irrational. Hence $3 - 2\sqrt{5}$ is irrational.

Chapter 2: Polynomials

YEAR: 2024

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For what value of k , the product of zeroes of the polynomial $kx^2 - 4x - 7$ is 2?

OPTIONS: (A) $1/14$ (B) $7/2$ (C) $7/2$ (D) $-2/7$

ANSWER: (D) $-2/7$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If one of the zeroes of the quadratic polynomial $(\alpha - 1)x^2 + \alpha x + 1$ is -3, then the value of α is:

OPTIONS: (A) $-2/3$ (B) $2/3$ (C) $4/3$ (D) $3/4$

ANSWER: (C) $4/3$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is:

OPTIONS: (A) $x^2 + 5x + 6$ (B) $x^2 - 5x + 6$ (C) $x^2 - 5x - 6$ (D) $-x^2 + 5x + 6$

ANSWER: (A) $x^2 + 5x + 6$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The zeroes of the polynomial $3x^2 + 11x - 4$ are:

OPTIONS: (A) $1/3, 4$ (B) $-1/3, -4$ (C) $1/3, -4$ (D) $-1/3, 4$

ANSWER: (C) $1/3, -4$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following equations has 2 as a root?

OPTIONS: (A) $x^2 - 4x + 5 = 0$ (B) $x^2 + 3x - 12 = 0$ (C) $2x^2 - 7x + 6 = 0$ (D) $3x^2 - 6x - 2 = 0$

ANSWER: (C) $2x^2 - 7x + 6 = 0$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The zeroes of the polynomial $3x^2 - 5x - 2$ are:

OPTIONS: (A) $1/3, 2$ (B) $-1/3, 2$ (C) $-1/3, -2$ (D) $1/3, -2$

ANSWER: (B) $-1/3, 2$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A quadratic polynomial having only zero (-2) is:

OPTIONS: (A) $(x - 2)^2$ (B) $x^2 - 2$ (C) $x^2 + 2x$ (D) $(x + 2)^2$

ANSWER: (D) $(x + 2)^2$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A quadratic polynomial whose zeroes are 3 and -2 , is:

OPTIONS: (A) $x^2 - x - 6$ (B) $x^2 + x - 6$ (C) $2x^2 - x - 12$ (D) $x^2 + x + 6$

ANSWER: (A) $x^2 - x - 6$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: One of the zeroes of the polynomial $p(x) = kx^2 - 9x + 3$ is $-3/2$. The value of k is:

OPTIONS: (A) $22/3$ (B) $-14/3$ (C) $14/3$ (D) $-22/3$

ANSWER: (D) $-22/3$

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: α, β are zeroes of the polynomial $2x^2 + 5x + 1$. The value of $(1/\alpha + 1/\beta)$ is:

OPTIONS: (A) $-5/4$ (B) 5 (C) $5/4$ (D) -5

ANSWER: (D) -5

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α, β are zeroes of the quadratic polynomial $2x^2 + 7x + 5$, find the value of $\alpha^2 + \beta^2 + \alpha\beta$.

ANSWER: $\alpha + \beta = -7/2$, $\alpha\beta = 5/2$. $\alpha^2 + \beta^2 + \alpha\beta = (\alpha + \beta)^2 - \alpha\beta = (-7/2)^2 - 5/2 = 49/4 - 10/4 = 39/4$

YEAR: 2024

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If one zero of the quadratic polynomial $6x^2 + 37x - (p - 2)$ is reciprocal of the other, find the value of p .

ANSWER: Product of zeroes = constant term/coefficient of $x^2 = -(p-2)/6$. If zeroes are reciprocal, product = 1. So $-(p-2)/6 = 1 \Rightarrow -(p-2) = 6 \Rightarrow p-2 = -6 \Rightarrow p = -4$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α, β are zeroes of the polynomial $8x^2 + 14x + 3$, find the value of $(1/\alpha + 1/\beta)$.

ANSWER: $\alpha + \beta = -14/8 = -7/4$, $\alpha\beta = 3/8$. $1/\alpha + 1/\beta = (\alpha + \beta)/\alpha\beta = (-7/4)/(3/8) = (-7/4) \times (8/3) = -14/3$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find a quadratic polynomial whose zeroes are -9 and 6.

ANSWER: Sum of zeroes = $-9 + 6 = -3$, Product = $-9 \times 6 = -54$. Polynomial = $x^2 - (\text{sum})x + \text{product} = x^2 + 3x - 54$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $p(x) = 3x^2 - 2x - 1$ and verify the relationship between the zeroes and the coefficients.

ANSWER: $p(x) = 3x^2 - 2x - 1 = (3x + 1)(x - 1)$. Zeroes are $x = -1/3, 1$. Sum of zeroes $= -1/3 + 1 = 2/3 = -\text{coefficient of } x / \text{coefficient of } x^2$. Product $= -1/3 \times 1 = -1/3 = \text{constant term} / \text{coefficient of } x^2$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $p(x) = 2x^2 + 5x + 2$ and verify the relationship between zeroes and its coefficients.

ANSWER: $p(x) = 2x^2 + 5x + 2 = (2x + 1)(x + 2)$. Zeroes are $x = -1/2, -2$. Sum of zeroes $= -1/2 + (-2) = -5/2 = -\text{coefficient of } x / \text{coefficient of } x^2$. Product $= -1/2 \times (-2) = 1 = \text{constant term} / \text{coefficient of } x^2$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $q(x) = 6x^2 - 5x - 1$ and verify the relationship between the zeroes and its coefficients.

ANSWER: $q(x) = 6x^2 - 5x - 1 = (6x + 1)(x - 1)$. Zeroes are $x = -1/6, 1$. Sum of zeroes $= -1/6 + 1 = 5/6 = -\text{coefficient of } x / \text{coefficient of } x^2$. Product $= -1/6 \times 1 = -1/6 = \text{constant term} / \text{coefficient of } x^2$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $4x^2 - 4x - 3$ and verify the relationship between zeroes and its coefficients.

ANSWER: $p(x) = 4x^2 - 4x - 3 = (2x + 1)(2x - 3)$. Zeroes are $x = -1/2, 3/2$. Sum $= -1/2 + 3/2 = 1 = -\text{coefficient of } x / \text{coefficient of } x^2$. Product $= -1/2 \times 3/2 = -3/4 = \text{constant term} / \text{coefficient of } x^2$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $4x^2 + 4x + 1$ and verify the relationship between the zeroes and coefficients.

ANSWER: $4x^2 + 4x + 1 = (2x + 1)^2$. Zeroes are $x = -1/2, -1/2$. Sum = $-1/2 + (-1/2) = -1 =$ -coefficient of x /coefficient of x^2 . Product = $(-1/2) \times (-1/2) = 1/4 =$ constant term/coefficient of x^2 .

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If zeroes of the quadratic polynomial $x^2 - 3x + 2$ are α and β , construct a quadratic polynomial whose zeroes are $2\alpha + 1$ and $2\beta + 1$.

ANSWER: $\alpha + \beta = 3, \alpha\beta = 2$. Sum of new zeroes = $2\alpha + 1 + 2\beta + 1 = 2(\alpha + \beta) + 2 = 2(3) + 2 = 8$. Product = $(2\alpha + 1)(2\beta + 1) = 4\alpha\beta + 2(\alpha + \beta) + 1 = 4(2) + 2(3) + 1 = 8 + 6 + 1 = 15$. Polynomial = $x^2 - 8x + 15$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If α, β are zeroes of the polynomial $3x^2 - 8x + 4$, form a quadratic polynomial in x whose zeroes are $1/\alpha$ and $1/\beta$.

ANSWER: $\alpha + \beta = 8/3, \alpha\beta = 4/3$. $1/\alpha + 1/\beta = (\alpha + \beta)/\alpha\beta = (8/3)/(4/3) = 2$. $1/(\alpha\beta) = 3/4$. Required polynomial = $x^2 - 2x + 3/4$ or $k(4x^2 - 8x + 3)$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If α, β are zeroes of the polynomial $8x^2 - 5x - 1$, form a quadratic polynomial in x whose zeroes are $2/\alpha$ and $2/\beta$.

ANSWER: $\alpha + \beta = 5/8, \alpha\beta = -1/8$. Sum = $2/\alpha + 2/\beta = 2(\alpha + \beta)/\alpha\beta = 2(5/8)/(-1/8) = -10$. Product = $(2/\alpha)(2/\beta) = 4/\alpha\beta = 4/(-1/8) = -32$. Required polynomial = $x^2 + 10x - 32$ or $k(x^2 + 10x - 32)$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find the zeroes of the polynomial $p(x) = 3x^2 + x - 10$ and verify the relationship between zeroes and its coefficients.

ANSWER: $p(x) = 3x^2 + x - 10 = (x + 2)(3x - 5)$. Zeroes are -2 and $5/3$. Sum = $-2 + 5/3 = -1/3 =$ -coefficient of x /coefficient of x^2 . Product = $-2 \times 5/3 = -10/3 =$ constant term/coefficient of x^2 .

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $9s^2 - 6s + 1$ and verify the relationship between the zeroes and the coefficients.

ANSWER: $9s^2 - 6s + 1 = (3s - 1)^2$. Zeroes are $1/3, 1/3$. Sum = $1/3 + 1/3 = 2/3 =$ -coefficient of s /coefficient of s^2 . Product = $1/3 \times 1/3 = 1/9 =$ constant term/coefficient of s^2 .

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $25a^2 - 10a + 1$ and verify the relationship between the zeroes and coefficients.

ANSWER: $25a^2 - 10a + 1 = (5a - 1)^2$. Zeroes are $1/5, 1/5$. Sum = $1/5 + 1/5 = 2/5 =$ -coefficient of a /coefficient of a^2 . Product = $1/5 \times 1/5 = 1/25 =$ constant term/coefficient of a^2 .

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If one zero of the polynomial $x^2 - 8x + k$ exceeds the other by 2, find the zeroes and the value of k .

ANSWER: Let zeroes be α and $\alpha + 2$. Sum = $2\alpha + 2 = 8 \Rightarrow \alpha = 3$. Zeroes are 3 and 5. Product = $3 \times 5 = 15 = k$.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find a quadratic polynomial, sum and product of whose zeroes are 5 and -6 respectively. Also, find the zeroes of the polynomial so obtained.

ANSWER: Polynomial = $x^2 - 5x - 6$. For zeroes: $x^2 - 5x - 6 = (x - 6)(x + 1)$, so zeroes are 6 and -1.

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CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Determine a quadratic polynomial, sum and product of whose zeroes are -10 and 24, respectively. Also, determine the zeroes of the polynomial so obtained.

ANSWER: Polynomial = $x^2 + 10x + 24$. For zeroes: $x^2 + 10x + 24 = (x + 4)(x + 6)$, so zeroes are -4 and -6.

Chapter 3: Pair of Linear Equations in Two Variables

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The lines represented by linear equations $x = a$ and $y = b$ ($a \neq b$) are:

OPTIONS: (A) intersecting at (a, b) (B) intersecting at (b, a) (C) parallel (D) coincident

ANSWER: (A) intersecting at (a, b)

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CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which the pair of linear equations $5x + 2y - 7 = 0$ and $2x + ky + 1 = 0$ don't have a solution, is:

OPTIONS: (A) 5 (B) $\frac{4}{5}$ (C) $\frac{5}{4}$ (D) $\frac{5}{2}$

ANSWER: (B) 4/5

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which the pair of linear equations $x + y - 4 = 0$ and $2x + ky - 8 = 0$ has infinitely many solutions, is:

OPTIONS: (A) $k \neq 2$ (B) $k \neq -2$ (C) $k = 2$ (D) $k = -2$

ANSWER: (C) $k = 2$

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The graph of a pair of linear equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ in two variables x and y represents parallel lines, if:

OPTIONS: (A) $a_1/a_2 \neq b_1/b_2$ (B) $a_1/a_2 = b_1/b_2 = c_1/c_2$ (C) $a_1/a_2 \neq b_1/b_2 = c_1/c_2$ (D) $a_1/a_2 = b_1/b_2 \neq c_1/c_2$

ANSWER: (D) $a_1/a_2 = b_1/b_2 \neq c_1/c_2$

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of solutions of the system of equations $x = 3$, $y = -1$ is:

OPTIONS: (A) 0 (B) 1 (C) 2 (D) Infinite

ANSWER: (B) 1

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The system of linear equations given by $x = a$ and $y = b$ is:

OPTIONS: (A) Consistent with a unique solution (B) Consistent with infinitely many solutions (C) Consistent with two solutions (D) Inconsistent

ANSWER: (A) Consistent with a unique solution

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve the following pair of linear equations algebraically: $x + 2y = 9$ and $y - 2x = 2$

ANSWER: From $y = 2x + 2$, substitute in $x + 2(2x + 2) = 9 \Rightarrow x + 4x + 4 = 9 \Rightarrow 5x = 5 \Rightarrow x = 1$, $y = 4$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Check whether the point $(-4, 3)$ lies on both the lines represented by the linear equations $x + y + 1 = 0$ and $x - y = 1$.

ANSWER: For $(-4, 3)$: $x + y + 1 = -4 + 3 + 1 = 0$, so it lies on first line. $x - y = -4 - 3 = -7 \neq 1$, so it does not lie on second line.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y : $3x + 5y = 8$, $5x - 3y = 2$

ANSWER: Multiplying first by 3 and second by 5: $9x + 15y = 24$, $25x - 15y = 10$. Adding: $34x = 34 \Rightarrow x = 1$. Substituting: $3 + 5y = 8 \Rightarrow 5y = 5 \Rightarrow y = 1$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y : $0.1x + 0.3y = 1$, $0.2x - 0.1y = -0.1$

ANSWER: Multiplying first by 10: $x + 3y = 10$. Second by 10: $2x - y = -1$. From second: $y = 2x + 1$. Substituting: $x + 3(2x + 1) = 10 \Rightarrow x + 6x + 3 = 10 \Rightarrow 7x = 7 \Rightarrow x = 1$, $y = 3$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the value of c for which the following pair of linear equations has infinitely many solutions: $cx + 3y = c - 3$, $12x + cy = c$

ANSWER: For infinitely many solutions: $c/12 = 3/c = (c-3)/c$. From first two: $c^2 = 36 \Rightarrow c = \pm 6$. From second and third: $3/c = (c-3)/c \Rightarrow 3 = c-3 \Rightarrow c = 6$. Therefore $c = 6$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y : $3x + 2y = 65$, $2x + 3y = 60$

ANSWER: Adding: $5x + 5y = 125 \Rightarrow x + y = 25$. Subtracting: $x - y = 5$. Solving: $x = 15$, $y = 10$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Solve graphically: $x + 3y = 6$, $2x - 3y = 12$

ANSWER: $x + 3y = 6 \Rightarrow x = 6 - 3y$. $2x - 3y = 12 \Rightarrow x = (12 + 3y)/2$. Solving: $6 - 3y = (12 + 3y)/2 \Rightarrow 12 - 6y = 12 + 3y \Rightarrow -9y = 0 \Rightarrow y = 0$, $x = 6$. Solution is $(6, 0)$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: x and y are complementary angles such that $x:y = 1:2$. Express the given information as a system of linear equations in two variables and hence solve it.

ANSWER: $x + y = 90^\circ$, $x/y = 1/2 \Rightarrow 2x = y$. Substitute: $x + 2x = 90 \Rightarrow 3x = 90 \Rightarrow x = 30^\circ$, $y = 60^\circ$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction.

ANSWER: Let fraction be $\frac{x}{y}$. $\frac{(x-1)}{y} = \frac{1}{3} \Rightarrow 3x - y = 3 \dots(i)$. $\frac{x}{(y+8)} = \frac{1}{4} \Rightarrow 4x - y = 8 \dots(ii)$.

Solving: $x = 5$, $y = 12$. Fraction = $\frac{5}{12}$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the value of k for which the following pair of linear equations will have infinitely many solutions: $kx + 3y - (k-3) = 0$ and $12x + ky - k = 0$. Hence find any two solutions.

ANSWER: For infinitely many solutions: $\frac{k}{12} = \frac{3}{k} = \frac{(k-3)}{k}$. From first two: $k^2 = 36 \Rightarrow k = \pm 6$.

From second and third: $\frac{3}{k} = \frac{(k-3)}{k} \Rightarrow 3 = k-3 \Rightarrow k = 6$. Equations: $6x + 3y = 3$ and $12x + 6y = 6$. Solutions: $(0, 1)$, $(\frac{1}{2}, 0)$, etc.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Solve graphically: $2x - y = 2$ and $4x - y = 4$. Also, write the coordinates of the points where the lines cut the y -axis.

ANSWER: For $2x - y = 2$: $y = 2x - 2$. For $4x - y = 4$: $y = 4x - 4$. Solving: $2x - 2 = 4x - 4 \Rightarrow -2x = -2 \Rightarrow x = 1$, $y = 0$. Lines cut y -axis at $(0, -2)$ and $(0, -4)$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: An academy offering cricket coaching bought 10 bats and 5 balls for ₹32,500.

Later, the academy bought 2 bats and 8 balls for ₹10,000. If there is no change in the cost of the bat and of the ball, find the cost of 1 bat and 1 ball.

ANSWER: Let cost of 1 bat = x , 1 ball = y . $10x + 5y = 32500 \Rightarrow 2x + y = 6500 \dots(i)$. $2x + 8y = 10000 \dots(ii)$. From (i): $y = 6500 - 2x$. Substitute in (ii): $2x + 8(6500 - 2x) = 10000 \Rightarrow 2x + 52000 - 16x = 10000 \Rightarrow -14x = -42000 \Rightarrow x = 3000$, $y = 500$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Solve graphically: $x + 2y = 10$ and $y = 3$

ANSWER: For $x + 2y = 10$, when $y = 3$: $x + 6 = 10 \Rightarrow x = 4$. Solution is (4, 3).

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Solve graphically: $2x + 3y = 12$ and $x - y = 1$

ANSWER: For $2x + 3y = 12$: $y = (12-2x)/3$. For $x - y = 1$: $y = x - 1$. Solving: $x - 1 = (12-2x)/3 \Rightarrow 3x - 3 = 12 - 2x \Rightarrow 5x = 15 \Rightarrow x = 3, y = 2$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Solve graphically: $2x + y = 9$ and $x - 2y = 2$

ANSWER: For $2x + y = 9$: $y = 9 - 2x$. For $x - 2y = 2$: $y = (x-2)/2$. Solving: $9 - 2x = (x-2)/2 \Rightarrow 18 - 4x = x - 2 \Rightarrow -5x = -20 \Rightarrow x = 4, y = 1$.

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Graphically solve: $3x - 4y + 3 = 0$ and $-2x + 5y = 9$

ANSWER: For $3x - 4y + 3 = 0$: $y = (3x+3)/4$. For $-2x + 5y = 9$: $y = (2x+9)/5$. Solving: $(3x+3)/4 = (2x+9)/5 \Rightarrow 5(3x+3) = 4(2x+9) \Rightarrow 15x + 15 = 8x + 36 \Rightarrow 7x = 21 \Rightarrow x = 3, y = 3$.

Chapter 4: Quadratic Equations

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the roots of quadratic equation $4x^2 - 5x + k = 0$ are real and equal, then value of k is:

OPTIONS: (A) $5/4$ (B) $25/16$ (C) $-5/4$ (D) $-25/16$

ANSWER: (B) $25/16$

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The discriminant of the quadratic equation $x^2 - 3x - 2 = 0$ is:

OPTIONS: (A) 1 (B) 17 (C) $\sqrt{17}$ (D) $-\sqrt{17}$

ANSWER: (B) 17

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The equation $x + 1/x = 3$ ($x \neq 0$) is expressed as a quadratic equation in the form of $ax^2 + bx + c = 0$. The value of $a - b + c$ is:

OPTIONS: (A) 5 (B) 2 (C) 1 (D) -1

ANSWER: (A) 5

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The roots of the quadratic equation $x^2 - 25 = 0$ is/are:

OPTIONS: (A) 5 (B) -5, 5 (C) 25 (D) -25, 25

ANSWER: (B) -5, 5

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The roots of the equation $x^2 - 8 = 0$ are:

OPTIONS: (A) rational and distinct (B) irrational and distinct (C) real and equal (D) not real

ANSWER: (B) irrational and distinct

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is not a quadratic equation?

OPTIONS: (A) $(x-2)^2 + 1 = 2x - 3$ (B) $(2x-1)(x-3) = (x+5)(x-1)$ (C) $x(x+1) + 8 = (x+2)(x-2)$ (D) $2x + \frac{3}{x} = 5$

ANSWER: (C) $x(x+1) + 8 = (x+2)(x-2)$

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following equations is a quadratic equation?

OPTIONS: (A) $x^2 = (x+1)^2$ (B) $(x-1)(x+2) = 2x + 1$ (C) $(x+2)^3 = 2x(x^2-1)$ (D) $\sqrt{x} = x^2$

ANSWER: (B) $(x-1)(x+2) = 2x + 1$

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of a for which $ax^2 + 3x + 1 = 0$ has real and equal roots is:

OPTIONS: (A) $\frac{4}{9}$ (B) $\frac{9}{4}$ (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

ANSWER: (B) $\frac{9}{4}$

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The discriminant of the quadratic equation $x^2 - 4x + 3 = 0$ is:

OPTIONS: (A) 28 (B) -8 (C) 4 (D) 2

ANSWER: (C) 4

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following quadratic equations has -1 as a root?

OPTIONS: (A) $x^2 - 4x - 5 = 0$ (B) $-x^2 - 4x + 5 = 0$ (C) $x^2 + 3x + 4 = 0$ (D) $x^2 - 5x + 6 = 0$

ANSWER: (A) $x^2 - 4x - 5 = 0$

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $1/2$ is a root of the quadratic equation $x^2 + kx - 5/4 = 0$, find the value of k .

ANSWER: Substitute $x = 1/2$: $(1/2)^2 + k(1/2) - 5/4 = 0 \Rightarrow 1/4 + k/2 - 5/4 = 0 \Rightarrow k/2 = 1 \Rightarrow k = 2$.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $2/3$ is a root of the quadratic equation $kx^2 - x - 2 = 0$, find the value of k .

ANSWER: Substitute $x = 2/3$: $k(4/9) - 2/3 - 2 = 0 \Rightarrow 4k/9 - 8/3 = 0 \Rightarrow 4k/9 = 8/3 \Rightarrow 4k = 24 \Rightarrow k = 6$.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Using quadratic formula, find the real roots of the equation $2x^2 + 2x + 9 = 0$, if they exist.

ANSWER: Discriminant $= 4 - 72 = -68 < 0$. Therefore no real roots exist.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the values of k for which the quadratic equation $kx^2 - 2kx + 6 = 0$ has real and equal roots. Also, find the roots.

ANSWER: For equal roots: $(-2k)^2 - 4(k)(6) = 0 \Rightarrow 4k^2 - 24k = 0 \Rightarrow 4k(k-6) = 0 \Rightarrow k = 0$ or $k = 6$.
 $k = 0$ gives equation $6 = 0$, not quadratic. So $k = 6$. Equation: $6x^2 - 12x + 6 = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow x = 1, 1$.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Express the equation $1/x - 1/(x-2) = 3$ as a quadratic equation in standard form. Hence find the roots.

ANSWER: $1/x - 1/(x-2) = 3 \Rightarrow (x-2-x)/(x(x-2)) = 3 \Rightarrow -2/(x^2-2x) = 3 \Rightarrow -2 = 3x^2 - 6x \Rightarrow 3x^2 - 6x + 2 = 0$. Discriminant $= 36 - 24 = 12$. Roots $= (6 \pm \sqrt{12})/6 = (6 \pm 2\sqrt{3})/6 = 1 \pm \sqrt{3}/3$.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The difference of the squares of two positive numbers is 180. The square of the smaller number is 8 times the greater number. Find the two numbers.

ANSWER: Let smaller $= y$, greater $= x$. $x^2 - y^2 = 180$, $y^2 = 8x$. $x^2 - 8x = 180 \Rightarrow x^2 - 8x - 180 = 0 \Rightarrow (x-18)(x+10) = 0 \Rightarrow x = 18$ (reject -10). $y^2 = 144 \Rightarrow y = 12$. Numbers are 18 and 12.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find the value(s) of k for which the equation $2x^2 + kx + 3 = 0$ has real and equal roots. Hence, find the roots of the equations so obtained.

ANSWER: For equal roots: $k^2 - 4(2)(3) = 0 \Rightarrow k^2 - 24 = 0 \Rightarrow k = \pm 2\sqrt{6}$. For $k = 2\sqrt{6}$: $2x^2 + 2\sqrt{6}x + 3 = 0 \Rightarrow$ roots $= -\sqrt{(3/2)}, -\sqrt{(3/2)}$. For $k = -2\sqrt{6}$: $2x^2 - 2\sqrt{6}x + 3 = 0 \Rightarrow$ roots $= \sqrt{(3/2)}, \sqrt{(3/2)}$.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The sum of areas of two squares is 2650 cm^2 . If the sum of their perimeters is 280 cm, find the sides of the two given squares.

ANSWER: Let sides be x and y . $x^2 + y^2 = 2650$, $4x + 4y = 280 \Rightarrow x + y = 70$. $(x+y)^2 = x^2 + 2xy + y^2 \Rightarrow 4900 = 2650 + 2xy \Rightarrow 2xy = 2250 \Rightarrow xy = 1125$. Numbers whose sum is 70 and product 1125 are 25 and 45. Sides are 25 cm and 45 cm.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find two consecutive odd integers, sum of whose squares is 290.

ANSWER: Let integers be x and $x+2$. $x^2 + (x+2)^2 = 290 \Rightarrow x^2 + x^2 + 4x + 4 = 290 \Rightarrow 2x^2 + 4x - 286 = 0 \Rightarrow x^2 + 2x - 143 = 0 \Rightarrow (x+13)(x-11) = 0 \Rightarrow x = 11$ (reject -13). Integers are 11 and 13.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A charity trust decides to build a rectangular hall having an area of 300 m^2 . The length of the hall is one metre more than twice its width. Find the length and breadth of the hall.

ANSWER: Let width = x , length = $2x+1$. $x(2x+1) = 300 \Rightarrow 2x^2 + x - 300 = 0 \Rightarrow (2x+25)(x-12) = 0 \Rightarrow x = 12$ (reject $-25/2$). Width = 12 m, Length = 25 m.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A rectangular field is 16 m long and 10 m wide. There is a path of equal width all around it, having an area of 120 sq.m . Find the width of the path.

ANSWER: Let width of path = x . $(16+2x)(10+2x) - 160 = 120 \Rightarrow 160 + 52x + 4x^2 - 160 = 120 \Rightarrow 4x^2 + 52x - 120 = 0 \Rightarrow x^2 + 13x - 30 = 0 \Rightarrow (x+15)(x-2) = 0 \Rightarrow x = 2$ (reject -15). Width = 2 m.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A cottage industry produces a certain number of toys in a day. The cost of production of each toy (in rupees) was found to be 55 minus the number of toys produced in a day. On a particular day, the total cost of production was ₹750. Find the total number of toys produced on that day.

ANSWER: Let number of toys = x . Cost per toy = $55 - x$. Total cost = $x(55-x) = 750 \Rightarrow 55x - x^2 = 750 \Rightarrow x^2 - 55x + 750 = 0 \Rightarrow (x-30)(x-25) = 0 \Rightarrow x = 30$ or $x = 25$.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A cottage industry produces a certain number of pottery articles in a day. It was observed that on a particular day the cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If the total cost of production on that day was ₹90, find the number of articles produced and the cost of each article.

ANSWER: Let number of articles = x . Cost per article = $2x + 3$. Total cost = $x(2x+3) = 90 \Rightarrow 2x^2 + 3x - 90 = 0 \Rightarrow (2x+15)(x-6) = 0 \Rightarrow x = 6$ (reject $-15/2$). Cost per article = $2(6)+3 = 15$.

Chapter 5: Arithmetic Progressions

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In an A.P., if $a = 8$ and $a_{10} = -19$, then value of d is:

OPTIONS: (A) 3 (B) $-11/9$ (C) $-27/10$ (D) -3

ANSWER: (D) -3

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The common difference of an A.P., if $a_{23} - a_{19} = 32$, is:

OPTIONS: (A) 8 (B) -8 (C) -4 (D) 4

ANSWER: (A) 8

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In an A.P., if $d = -4$ and $a_7 = 4$, then the first term a is equal to:

OPTIONS: (A) 6 (B) 7 (C) 20 (D) 28

ANSWER: (D) 28

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The 20th term of the A.P.: $10\sqrt{2}, 6\sqrt{2}, 2\sqrt{2}, \dots$ is:

OPTIONS: (A) $-76 + 10\sqrt{2}$ (B) $-62\sqrt{2}$ (C) $-66\sqrt{2}$ (D) $86\sqrt{2}$

ANSWER: (C) $-66\sqrt{2}$

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The 10th term of the A.P.: $-12, -19, -26, \dots$ is:

OPTIONS: (A) -75 (B) -65 (C) 51 (D) -82

ANSWER: (A) -75

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The 16th term of the A.P.: $5\sqrt{3}, 2\sqrt{3}, -\sqrt{3}, \dots$ is:

OPTIONS: (A) $-25\sqrt{3}$ (B) $-40\sqrt{3}$ (C) $50\sqrt{3}$ (D) $-45+5\sqrt{3}$

ANSWER: (B) $-40\sqrt{3}$

YEAR: 2024
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The 30th term of the A.P.: -3, -7, -11, ... is:
OPTIONS: (A) 113 (B) -117 (C) -119 (D) -120
ANSWER: (C) -119

YEAR: 2024
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The 22nd term of the A.P.: $\frac{3}{2}$, $\frac{1}{2}$, $-\frac{1}{2}$, $-\frac{3}{2}$, ... is:
OPTIONS: (A) $\frac{45}{2}$ (B) -9 (C) $-\frac{39}{2}$ (D) -21
ANSWER: (C) $-\frac{39}{2}$

YEAR: 2024
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The 15th term of the A.P.: $\frac{13}{3}$, $\frac{9}{3}$, $\frac{5}{3}$, ... is:
OPTIONS: (A) 23 (B) $-\frac{53}{3}$ (C) -11 (D) $-\frac{43}{3}$
ANSWER: (D) $-\frac{43}{3}$

YEAR: 2024
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Find the sum of the A.P. 7, $10\frac{1}{2}$, 14, ... 84.
ANSWER: $a = 7$, $d = \frac{7}{2}$. $84 = 7 + (n-1)(\frac{7}{2}) \Rightarrow 77 = (n-1)(\frac{7}{2}) \Rightarrow n-1 = 22 \Rightarrow n = 23$. $S_{23} = \frac{23}{2}(7+84) = \frac{23}{2} \times 91 = 2093\frac{1}{2}$.

YEAR: 2024
CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If the sum of first n terms of an A.P. is given by $S_n = n/2(2n+8)$, find its first term and common difference. Hence, find its 15th term.

ANSWER: $S_1 = a = 1/2(2+8) = 5$. $S_2 = 2/2(4+8) = 12$. $d = S_2 - 2a = 12 - 10 = 2$. $a_{15} = a + 14d = 5 + 28 = 33$.

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: How many terms of the A.P. 27, 24, 21, ... must be taken so that their sum is 105? Which term of the A.P. is zero?

ANSWER: $a = 27$, $d = -3$. $S_n = n/2[2(27) + (n-1)(-3)] = 105 \Rightarrow n/2[54 - 3n + 3] = 105 \Rightarrow n(57 - 3n) = 210 \Rightarrow 3n^2 - 57n + 210 = 0 \Rightarrow n^2 - 19n + 70 = 0 \Rightarrow (n-5)(n-14) = 0 \Rightarrow n = 5$ or 14 . Zero term: $27 + (n-1)(-3) = 0 \Rightarrow 27 - 3n + 3 = 0 \Rightarrow 30 - 3n = 0 \Rightarrow n = 10$.

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If the sum of the first 7 terms of an A.P. is 91 and that of the first 17 terms is 561, then find the sum of the first n terms and hence find the n th term.

ANSWER: $S_7 = 7/2[2a + 6d] = 91 \Rightarrow 2a + 6d = 26 \Rightarrow a + 3d = 13$. $S_{17} = 17/2[2a + 16d] = 561 \Rightarrow 2a + 16d = 66 \Rightarrow a + 8d = 33$. Solving: $5d = 20 \Rightarrow d = 4$, $a = 1$. $S_n = n/2[2(1) + (n-1)4] = n/2[2 + 4n - 4] = n(2n - 1)$. $a_n = 1 + (n-1)4 = 4n - 3$.

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In an A.P. if $S_n = 4n^2 - n$, find (i) first term and common difference (ii) the A.P. (iii) which term of the A.P. is 107?

ANSWER: $a = S_1 = 4(1)^2 - 1 = 3$. $a_2 = S_2 - S_1 = [4(4) - 2] - 3 = 16 - 2 - 3 = 11$. $d = 8$. A.P.: 3, 11, 19, 27... For 107: $3 + (n-1)8 = 107 \Rightarrow 8(n-1) = 104 \Rightarrow n-1 = 13 \Rightarrow n = 14$.

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: Sumant's mother started a new shoe shop. To display the shoes, she put 3 pairs of shoes in the 1st row, 5 pairs in the 2nd row, 7 pairs in the 3rd row and so on. (i) How many pairs of shoes are displayed in the 6th row? (ii) What is the difference of pairs of shoes in the 1st row and the 6th row? (iii)(a) Find the total number of pairs of shoes displayed in the first 15 rows. OR (b) If the pairs of shoes displayed in the 4th row are on sale at price of ₹500 for each pair, find the total amount earned if all shoes displayed in the 4th row are sold out.

ANSWER: (i) $a_6 = 3 + 5(3) = 18$. (ii) Difference = $18 - 3 = 15$. (iii)(a) $S_{15} = 15/2[2(3) + 14(3)] = 15/2[6+42] = 15/2 \times 48 = 360$. (b) $a_4 = 3 + 3(3) = 12$. Amount = $12 \times 500 = 6000$.

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: In a garden, saplings of rose flowers were planted at equal intervals to form a spiral pattern. The spiral is made up of successive semicircles, with radii 50 cm, 100 cm, 150 cm,... Spiral 1 has 10 flowers, Spiral 2 has 20 flowers, Spiral 3 has 30 flowers and so on. (i) What is the radius of the 13th spiral? (ii) If the radius of the nth spiral is 500 cm, find n. (iii)(a) Find the total number of saplings till the 11th spiral. OR (b) Till which spiral, will there be a total of 450 saplings?

ANSWER: (i) $a_{13} = 50 + 12(50) = 650$ cm. (ii) $500 = 50 + (n-1)50 \Rightarrow 450 = 50(n-1) \Rightarrow n-1 = 9 \Rightarrow n = 10$. (iii)(a) $S_{11} = 11/2[2(10) + 10(10)] = 11/2[20+100] = 11/2 \times 120 = 660$. (b) $450 = n/2[20 + (n-1)10] \Rightarrow 900 = n(10 + 10n) \Rightarrow 10n^2 + 10n - 900 = 0 \Rightarrow n^2 + n - 90 = 0 \Rightarrow (n+10)(n-9) = 0 \Rightarrow n = 9$.

YEAR: 2024

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: A woman borrowed ₹10,00,000 from her friend and promised to return the borrowed money in monthly instalments beginning from the next month. After one month, she returned ₹10,000, the next month ₹15,000, the third month ₹20,000 and so on. (i) Find the amount of instalment paid in the tenth month. (ii) In which instalment did she pay ₹40,000? (iii)(a) If she returned ₹11,50,000 in all, how many instalments did she pay? OR (b) By which instalment has she returned a total amount of ₹3,25,000?

ANSWER: (i) $a_{10} = 10000 + 9(5000) = 55000$. (ii) $40000 = 10000 + (n-1)5000 \Rightarrow 30000 = 5000(n-1) \Rightarrow n-1 = 6 \Rightarrow n = 7$. (iii)(a) $1150000 = n/2[2(10000) + (n-1)5000] \Rightarrow 2300000 =$

$n[20000 + 5000n - 5000] \Rightarrow 2300000 = n(15000 + 5000n) \Rightarrow 5000n^2 + 15000n - 2300000 = 0$
 $\Rightarrow n^2 + 3n - 460 = 0 \Rightarrow (n+23)(n-20) = 0 \Rightarrow n = 20$. (b) $325000 = n/2[20000 + (n-1)5000] \Rightarrow$
 $650000 = n[15000 + 5000n] \Rightarrow 5000n^2 + 15000n - 650000 = 0 \Rightarrow n^2 + 3n - 130 = 0 \Rightarrow$
 $(n+13)(n-10) = 0 \Rightarrow n = 10$.

Chapter 6: Coordinate Geometry

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance between the points (2, -3) and (-2, 3) is:

OPTIONS: (A) $2\sqrt{13}$ units (B) 5 units (C) $13\sqrt{2}$ units (D) 10 units

ANSWER: (A) $2\sqrt{13}$ units

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The diameter of a circle is of length 6 cm. If one end of the diameter is (-4, 0), the other end on x-axis is at:

OPTIONS: (A) (0, 2) (B) (6, 0) (C) (2, 0) (D) (4, 0)

ANSWER: (C) (2, 0)

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mid-point of the line segment joining the points (-1, 3) and (8, $\frac{3}{2}$) is:

OPTIONS: (A) ($\frac{7}{2}$, $-\frac{3}{4}$) (B) ($\frac{7}{2}$, $\frac{9}{2}$) (C) ($\frac{9}{2}$, $-\frac{3}{4}$) (D) ($\frac{7}{2}$, $\frac{9}{4}$)

ANSWER: (D) ($\frac{7}{2}$, $\frac{9}{4}$)

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of the point (2, 3) from the origin is:

OPTIONS: (A) 2 (B) 3 (C) 5 (D) $\sqrt{13}$

ANSWER: (D) $\sqrt{13}$

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of the point (3, 4) from the origin is:

OPTIONS: (A) 25 (B) 5 (C) 7 (D) 1

ANSWER: (B) 5

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of the point (5, 4) from the origin is:

OPTIONS: (A) 41 (B) $\sqrt{41}$ (C) 3 (D) 9

ANSWER: (B) $\sqrt{41}$

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of point (-3, 4) from y-axis is:

OPTIONS: (A) -3 (B) 3 (C) 4 (D) 5

ANSWER: (B) 3

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If point (1, 2) is the mid-point of the line segment joining the points (3, 5) and (2a, b), then (a, b) =

OPTIONS: (A) (-1, -1) (B) (-1/2, -1/2) (C) (-1/2, -1) (D) (-1, -1/2)

ANSWER: (C) (-1/2, -1)

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the ratio in which the point (3, y) divides the line segment joining the points (-2, -5) and (6, 3). Also, find the value of y.

ANSWER: Let ratio be k:1. $x = \frac{-2+6k}{k+1} = 3 \Rightarrow -2+6k = 3k+3 \Rightarrow 3k = 5 \Rightarrow k = \frac{5}{3}$. $y = \frac{-5+3k}{k+1} = \frac{-5+5}{(5/3+1)} = \frac{0}{(8/3)} = 0$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find a point on y-axis which is equidistant from the points A(6, 5) and B(-4, 3).

ANSWER: Let point be (0, y). Distance to A: $\sqrt{36 + (y-5)^2}$. Distance to B: $\sqrt{16 + (y-3)^2}$.

Equating: $36 + (y-5)^2 = 16 + (y-3)^2 \Rightarrow 36 + y^2 - 10y + 25 = 16 + y^2 - 6y + 9 \Rightarrow 61 - 10y = 25 - 6y \Rightarrow 36 = 4y \Rightarrow y = 9$. Point is (0, 9).

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If A(1,1) and B(7,9) are the end points of a diameter of a circle, find the coordinates of the centre of the circle.

ANSWER: Centre = midpoint = $\left(\frac{1+7}{2}, \frac{1+9}{2}\right) = (4, 5)$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the ratio in which the point P(-1, m) divides the line segment joining the points A(2, 5) and B(-5, -2). Hence, find the value of m.

ANSWER: Let ratio be $k:1$. $x = (2 - 5k)/(k+1) = -1 \Rightarrow 2 - 5k = -k - 1 \Rightarrow -4k = -3 \Rightarrow k = 3/4$. $m = (5 - 2k)/(k+1) = (5 - 1.5)/(1.75) = 3.5/1.75 = 2$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the coordinates of the points of trisection of the line segment joining the points $(-4, 1)$ and $(6, 5)$.

ANSWER: $P = (1/3)(6,5) + (2/3)(-4,1) = (6/3 - 8/3, 5/3 + 2/3) = (-2/3, 7/3)$. $Q = (2/3)(6,5) + (1/3)(-4,1) = (12/3 - 4/3, 10/3 + 1/3) = (8/3, 11/3)$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the coordinates of the points of trisection of the line segment joining the points $(-2, 2)$ and $(7, -4)$.

ANSWER: $P = (1/3)(7,-4) + (2/3)(-2,2) = (7/3 - 4/3, -4/3 + 4/3) = (1, 0)$. $Q = (2/3)(7,-4) + (1/3)(-2,2) = (14/3 - 2/3, -8/3 + 2/3) = (4, -2)$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The line AB intersects x-axis at A and y-axis at B. The point $P(2, -3)$ lies on AB such that $AP:PB = 3:1$. Find the coordinates of A and B.

ANSWER: Let $A = (a, 0)$, $B = (0, b)$. P divides AB in ratio 3:1. $P = (1/4)A + (3/4)B$? Wait $AP:PB = 3:1$ means P divides AB in ratio 3:1 from A. $P = (1/4)A + (3/4)B$? Actually if $AP:PB = 3:1$, then $P = (1/4)A + (3/4)B$. So $(2, -3) = (a/4, 3b/4) \Rightarrow a = 8, b = -4$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If points A(-5, y), B(2, -2), C(8, 4) and D(x, 5) taken in order form a parallelogram ABCD, find the values of x and y. Hence, find lengths of sides of the parallelogram.

ANSWER: Midpoint of AC = $(-5+8)/2, (y+4)/2 = (3/2, (y+4)/2)$. Midpoint of BD = $(2+x)/2, (-2+5)/2 = ((2+x)/2, 3/2)$. Equating: $(2+x)/2 = 3/2 \Rightarrow x = 1$. $(y+4)/2 = 3/2 \Rightarrow y = -1$. $AB = \sqrt{((2+5)^2 + (-2+1)^2)} = \sqrt{(49+1)} = \sqrt{50} = 5\sqrt{2}$. $BC = \sqrt{((8-2)^2 + (4+2)^2)} = \sqrt{(36+36)} = \sqrt{72} = 6\sqrt{2}$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A(6,-3), B(0,5) and C(-2,1) are vertices of ΔABC . Points P(3,1) and Q(2,-1) lie on sides AB and AC respectively. Check whether $AP/PB = AQ/QC$.

ANSWER: $AP = \sqrt{((3-6)^2 + (1+3)^2)} = \sqrt{(9+16)} = 5$. $PB = \sqrt{((3-0)^2 + (1-5)^2)} = \sqrt{(9+16)} = 5$. $AP/PB = 1$. $AQ = \sqrt{((2-6)^2 + (-1+3)^2)} = \sqrt{(16+4)} = \sqrt{20} = 2\sqrt{5}$. $QC = \sqrt{((2+2)^2 + (-1-1)^2)} = \sqrt{(16+4)} = \sqrt{20} = 2\sqrt{5}$. $AQ/QC = 1$. Therefore $AP/PB = AQ/QC$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: In a society, there is a circular park having two gates at points A(10,20) and B(50,50). Two fountains are installed at points P and Q on AB such that $AP = PQ = QB$. (i) Find the coordinates of the centre C. (ii) Find the radius of the circular park. (iii)(a) Find the coordinates of point P. OR (b) Find the distance of the fountain at Q from gate A.

ANSWER: (i) $C = ((10+50)/2, (20+50)/2) = (30, 35)$. (ii) Radius = $\sqrt{((30-10)^2 + (35-20)^2)} = \sqrt{(400+225)} = \sqrt{625} = 25$. (iii)(a) P divides AB in ratio 1:2. $P = (1/3)(50,50) + (2/3)(10,20) = (50/3+20/3, 50/3+40/3) = (70/3, 90/3) = (70/3, 30)$. (b) $AB = 2 \times 25 = 50$. $AQ = 2/3 \times 50 = 100/3$.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: A field is in the form of a rectangle. Coordinates of ABCD are A(10,10), B(40,10), C(40,50) and D(x,y). Anita started from A to E along diagonal AC, where E is the intersection of diagonals. From E to B along DB and back to A along BA. (i) Find coordinates of E. (ii) Find distance between B and C. (iii)(a) Find coordinates of D and distance BD. OR (b) Find total distance travelled by Anita.

ANSWER: (i) E = midpoint of AC = (25, 30). (ii) $BC = \sqrt{(40-40)^2 + (50-10)^2} = 40$. (iii)(a) D = (10, 50). $BD = \sqrt{(40-10)^2 + (10-50)^2} = \sqrt{900+1600} = 50$. (b) $AE = \sqrt{(25-10)^2 + (30-10)^2} = \sqrt{225+400} = 25$. $EB = \sqrt{(40-25)^2 + (10-30)^2} = \sqrt{225+400} = 25$. $BA = \sqrt{(40-10)^2 + (10-10)^2} = 30$. Total = 80.

Chapter 7: Triangles

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is not the criterion for similarity of triangles?

OPTIONS: (A) AAA (B) SSS (C) SAS (D) RHS

ANSWER: None of the given options is correct. (All are criteria for similarity)

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: What values of x and y will make $\triangle ABC$ similar to $\triangle QRP$?

OPTIONS: (A) $x = 6, y = 5$ (B) $x = 5, y = 6$ (C) $x = 6, y = 6$ (D) $x = 12, y = 3/2$

ANSWER: (B) $x = 5, y = 6$

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which types of triangles are always similar?

OPTIONS: (A) Right-angled triangles (B) Acute-angled triangles (C) Isosceles triangles (D) Equilateral triangles

ANSWER: (D) Equilateral triangles

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: D is a point on side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Show that $AC^2 = BC \times DC$.

ANSWER: In $\triangle ADC$ and $\triangle BAC$: $\angle ADC = \angle BAC$ (given), $\angle C = \angle C$ (common). Therefore $\triangle ADC \sim \triangle BAC$ (AA similarity). So $AC/BC = DC/AC \Rightarrow AC^2 = BC \times DC$.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, $AB \parallel DE$ and $BD \parallel EF$. Prove that $DC^2 = CF \times AC$.

ANSWER: $EF \parallel BD \Rightarrow CF/DC = CE/CB$. $DE \parallel AB \Rightarrow DC/AC = CE/CB$. Therefore $CF/DC = DC/AC \Rightarrow DC^2 = CF \times AC$.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: State Basic Proportionality Theorem and use it to prove: A line through the mid-point of one side of a triangle, parallel to another side, bisects the third side.

ANSWER: BPT: If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Given: In $\triangle ABC$, P is mid-point of AB and $PQ \parallel BC$. To prove: Q is mid-point of AC. Since $PQ \parallel BC$, $AP/PB = AQ/QC$. $AP = PB \Rightarrow AP/PB = 1 \Rightarrow AQ/QC = 1 \Rightarrow AQ = QC$.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: State Basic Proportionality Theorem and use it to prove: In a quadrilateral ABCD, diagonals AC and BD intersect at O such that $AO/BO = CO/DO$. Prove that ABCD is a trapezium.

ANSWER: BPT: If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Construction: Draw $OE \parallel AB$. In $\triangle DAB$, $OE \parallel AB \Rightarrow DE/AE = DO/BO$. Given $AO/BO = CO/DO \Rightarrow DO/BO = CO/AO$. So $DE/AE = CO/AO$. By converse of BPT, $OE \parallel CD$. Since $OE \parallel AB$ and $OE \parallel CD$, $AB \parallel CD$. Hence ABCD is a trapezium.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: State the converse of Basic Proportionality Theorem and use it to prove: Line segment joining mid-points of any two sides of a triangle is parallel to the third side.

ANSWER: Converse of BPT: If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. Given: D and E are mid-points of AB and AC. $AD/DB = 1$, $AE/EC = 1 \Rightarrow AD/DB = AE/EC$. By converse of BPT, $DE \parallel BC$.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Sides AB and BC and median AD of $\triangle ABC$ are respectively proportional to sides PQ and QR and median PM of $\triangle PQR$. Show that $\triangle ABC \sim \triangle PQR$.

ANSWER: Given $AB/PQ = BC/QR = AD/PM$. Extend AD to E such that $AD = DE$ and PM to N such that $PM = MN$. Then $AB = CE$ (by congruence), $PQ = RN$. So $CE/RN = AB/PQ = AD/PM$. Also $AE = 2AD$, $PN = 2PM \Rightarrow AE/PN = AD/PM = AB/PQ$. Thus $CE/RN = AE/PN = AC/PR$. Therefore $\triangle ACE \sim \triangle PRN$ (SSS), so $\angle CAE = \angle RPN$. Similarly $\angle BAD = \angle QPM$. Adding: $\angle BAC = \angle QPR$. Hence $\triangle ABC \sim \triangle PQR$.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, $\angle ABC = \angle ACB$ and $BC/BE = BD/AC$. Show that $\triangle ABE \sim \triangle DBC$ and $AE \parallel DC$.

ANSWER: Given $\angle ABC = \angle ACB \Rightarrow AB = AC$. $BC/BE = BD/AC \Rightarrow BC/BE = BD/AB$. So $BC/BD = BE/AB$. Also $\angle ABC = \angle DBC$ (common). Therefore $\triangle ABE \sim \triangle DBC$ (SAS similarity). Thus $\angle BAE = \angle BDC$. Since $\angle BAE$ and $\angle BDC$ are corresponding angles, $AE \parallel DC$.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If BD and QM are medians of triangles ABC and PQR respectively, where $\triangle ABC \sim \triangle PQR$, prove that $AB/PQ = BD/QM$.

ANSWER: Given $\triangle ABC \sim \triangle PQR \Rightarrow AB/PQ = BC/QR = AC/PR$ and $\angle B = \angle Q$. Since BD and QM are medians, D is midpoint of AC and M is midpoint of PR. So $DC/PR = (AC/2)/PR = AC/(2PR) = AB/(2PQ)$. Also $BC/QR = AB/PQ$. Therefore $BC/QR = DC/PR$ and $\angle C = \angle R$. Thus $\triangle BCD \sim \triangle QRM$ (SAS), so $BD/QM = BC/QR = AB/PQ$.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, prove that the other two sides are divided in the same ratio.

ANSWER: Given: In $\triangle ABC$, $DE \parallel BC$. To prove: $AD/DB = AE/EC$. Construction: Join BE, CD. Draw $DM \perp AC$, $EN \perp AB$. $\text{ar}(\triangle ADE)/\text{ar}(\triangle BDE) = (1/2 \times AD \times EN)/(1/2 \times DB \times EN) = AD/DB$. $\text{ar}(\triangle ADE)/\text{ar}(\triangle CDE) = (1/2 \times AE \times DM)/(1/2 \times EC \times DM) = AE/EC$. Since $\triangle BDE$ and $\triangle CDE$ lie on same base DE and between parallels DE and BC, $\text{ar}(\triangle BDE) = \text{ar}(\triangle CDE)$. Hence $AD/DB = AE/EC$.

Chapter 8: Circles

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PA is a tangent to a circle with centre O. If $OP = 10$ cm, then the length of AP is:

OPTIONS: (A) $10\sqrt{3}$ cm (B) 20 cm (C) 5 cm (D) $5\sqrt{3}$ cm

ANSWER: (D) $5\sqrt{3}$ cm

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In two concentric circles, a tangent to the smaller circle will intersect the larger circle at:

OPTIONS: (A) zero point (B) one point (C) two points (D) three points

ANSWER: (C) two points

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: PQ and PR are tangents to the circle of radius 3 cm and centre O. If length of each tangent is 4 cm, then perimeter of ΔOQP is:

OPTIONS: (A) 5 cm (B) 12 cm (C) 9 cm (D) 8 cm

ANSWER: (B) 12 cm

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: PA and PB are tangents to a circle with centre O. If $\angle AOB = 105^\circ$ then $\angle OAP + \angle APB$ is equal to:

OPTIONS: (A) 75° (B) 175° (C) 180° (D) 165°

ANSWER: (D) 165°

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: PQ and PR are tangents to a circle with centre O such that $OQ = QP$. The value of $\angle OPQ$ is equal to:

OPTIONS: (A) 45° (B) 30° (C) 60° (D) 90°

ANSWER: (A) 45°

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A line intersecting a circle in two distinct points is called a:

OPTIONS: (A) secant (B) chord (C) diameter (D) tangent

ANSWER: (A) secant

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A tangent to a circle is a line that touches the circle at:

OPTIONS: (A) one point only (B) two points (C) three points (D) infinite number of points

ANSWER: (A) one point only

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: How many tangents can be drawn from point P on the outer circle to the inner circle in the given figure?

OPTIONS: (A) 0 (B) 1 (C) 2 (D) Infinite

ANSWER: (C) 2

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: XY and PQ are two tangents drawn at the end points of the diameter AB of a circle. Prove that $XY \parallel PQ$.

ANSWER: Since tangent is perpendicular to radius at point of contact, $XY \perp OA$ and $PQ \perp OB$. Since OA and OB are collinear (diameter), XY and PQ are both perpendicular to the same line, hence they are parallel.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Prove that the tangent drawn at any point of the circle is perpendicular to the radius through the point of contact.

ANSWER: Given: OA is radius and ℓ is tangent at A . Let B be any point on ℓ other than A . B lies outside the circle, so $OB > \text{radius} = OA$. Since perpendicular distance is the shortest distance, $OA \perp \ell$.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, TP and TQ are two tangents. If $\angle PTQ = 50^\circ$, find the measure of $\angle OPQ$.

ANSWER: $TP = TQ$ (tangents from external point). $\angle TPQ = \angle TQP$. In $\triangle TPQ$: $50^\circ + 2\angle TPQ = 180^\circ \Rightarrow \angle TPQ = 65^\circ$. $\angle OPQ = 90^\circ - 65^\circ = 25^\circ$.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that a rectangle circumscribing a circle is a square.

ANSWER: As lengths of tangents from an external point to a circle are equal: $AP = AS$, $BP = BQ$, $DR = DS$, $CR = CQ$. Adding: $AB + CD = BC + AD$. As opposite sides of rectangle are equal, $AB = CD$ and $BC = AD \Rightarrow 2AB = 2AD \Rightarrow AB = AD$. Therefore rectangle is a square.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that the parallelogram circumscribing a circle is a rhombus.

ANSWER: As lengths of tangents from an external point to a circle are equal: $AP = AS$, $BP = BQ$, $DR = DS$, $CR = CQ$. Adding: $AB + CD = BC + AD$. As opposite sides of parallelogram are equal, $AB = CD$ and $BC = AD \Rightarrow 2AB = 2AD \Rightarrow AB = AD$. Therefore parallelogram is a rhombus.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre.

ANSWER: Since tangent is perpendicular to radius at point of contact, $\angle OAP = \angle OBP = 90^\circ$. In quadrilateral OAPB, $\angle OAP + \angle OBP + \angle APB + \angle AOB = 360^\circ \Rightarrow 90^\circ + 90^\circ + \angle APB + \angle AOB = 360^\circ \Rightarrow \angle APB + \angle AOB = 180^\circ$. Hence supplementary.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Prove that the lengths of the tangents drawn from an external point to a circle are equal.

ANSWER: Given: PA and PB are tangents from P to circle with centre O. In $\triangle OAP$ and $\triangle OBP$:
 $OA = OB$ (radii), $\angle OAP = \angle OBP = 90^\circ$, $OP = OP$ (common). Therefore $\triangle OAP \cong \triangle OBP$ (RHS).
Hence $PA = PB$.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Two chords BA and CD intersect at point P outside the circle. Prove that $\triangle PDA \sim \triangle PBC$.

ANSWER: $\angle PDA = \angle PBC$ (angles in same segment). $\angle P$ is common. Therefore $\triangle PDA \sim \triangle PBC$ (AA similarity).

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, TP and TQ are tangents at points P and Q of the circle respectively. If reflex $\angle POQ = 250^\circ$, find the measure of each angle of quadrilateral POQT.

ANSWER: $\angle POQ = 360^\circ - 250^\circ = 110^\circ$. $\angle OPT = \angle OQT = 90^\circ$ (tangent perpendicular to radius). In quadrilateral OPTQ: $\angle OPT + \angle OQT + \angle POQ + \angle PTQ = 360^\circ \Rightarrow 90^\circ + 90^\circ + 110^\circ + \angle PTQ = 360^\circ \Rightarrow \angle PTQ = 70^\circ$.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, O is the centre of a circle of radius 7 cm. AB is a chord of the circle. Find the perimeter of the shaded region. (Given figure has angle 60°)

ANSWER: $\triangle AOB$ is equilateral as $\angle AOB = 60^\circ$, so $AB = 7$ cm. Length of minor arc $AB = \frac{60}{360} \times 2 \times \frac{22}{7} \times 7 = \frac{22}{3}$ cm. Perimeter = $\frac{22}{3} + 7 = \frac{43}{3}$ cm.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A quadrilateral circumscribes the circle. If $AB = 5$ cm, $BC = 7$ cm and $CD = 6$ cm, find the length of AD .

ANSWER: For a quadrilateral circumscribing a circle, $AB + CD = BC + AD \Rightarrow 5 + 6 = 7 + AD \Rightarrow AD = 4$ cm.

Chapter 9: Introduction to Trigonometry

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sin \theta = \frac{1}{3}$, then $\sec \theta$ is equal to:

OPTIONS: (A) $2\sqrt{2}/3$ (B) $3/2\sqrt{2}$ (C) 3 (D) $1/\sqrt{3}$

ANSWER: (B) $3/2\sqrt{2}$

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For what value of θ , $\sin^2\theta + \sin\theta + \cos^2\theta$ is equal to 2?

OPTIONS: (A) 45° (B) 0° (C) 90° (D) 30°

ANSWER: (A) 45°

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following statements is false?

OPTIONS: (A) $\tan 45^\circ = \cot 45^\circ$ (B) $\sin 90^\circ = \tan 45^\circ$ (C) $\sin 30^\circ = \cos 30^\circ$ (D) $\sin 45^\circ = \cos 45^\circ$

ANSWER: (C) $\sin 30^\circ = \cos 30^\circ$

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(\tan^2 A - 1/\cos^2 A)$ is:

OPTIONS: (A) more than 1 (B) 1 (C) 0 (D) -1

ANSWER: (D) -1

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sin A = 3/5$, then $\cot A$ is equal to:

OPTIONS: (A) $3/4$ (B) $4/3$ (C) $4/5$ (D) $5/4$

ANSWER: (B) $4/3$

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sin A = 2/3$, then $\cos A$ is equal to:

OPTIONS: (A) $3/2$ (B) $\sqrt{5}/3$ (C) $1/3$ (D) $1/\sqrt{3}$

ANSWER: (B) $\sqrt{5}/3$

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\tan A = 1$, then $3 \sin A + \cos A$ is equal to:

OPTIONS: (A) $4\sqrt{2}$ (B) 4 (C) $2\sqrt{2}$ (D) $4 \times 45^\circ$

ANSWER: (C) $2\sqrt{2}$

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(\cot^2 A - 1/\sin^2 A)$ is:

OPTIONS: (A) more than 1 (B) 1 (C) 0 (D) -1

ANSWER: (D) -1

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ - \cot 45^\circ$

ANSWER: $\sin 30^\circ \cos 60^\circ = 1/2 \times 1/2 = 1/4$. $\cos 30^\circ \sin 60^\circ = \sqrt{3}/2 \times \sqrt{3}/2 = 3/4$. Sum = 1. $\cot 45^\circ = 1$. Result = $1 - 1 = 0$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the values of A and B if $\tan(A+B) = 1$ and $\tan(A-B) = 1/\sqrt{3}$.

ANSWER: $A+B = 45^\circ$, $A-B = 30^\circ$. Adding: $2A = 75^\circ \Rightarrow A = 37.5^\circ$, $B = 7.5^\circ$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Prove that $\tan 45^\circ = 1$ geometrically.

ANSWER: Consider an isosceles right $\triangle ABC$ with $AB = BC = x$. Using angle sum property, $\angle A = \angle C = 45^\circ$. $\tan 45^\circ = AB/BC = x/x = 1$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $\sin 3A = 1$, find the value of $\cos 2A - \tan^2 45^\circ$.

ANSWER: $3A = 90^\circ \Rightarrow A = 30^\circ$. $\cos 60^\circ - 1 = 1/2 - 1 = -1/2$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $(\sec A + \tan A)(1 - \sin A) = k \cos A$, find the value of k .

ANSWER: $(\sec A + \tan A)(1 - \sin A) = (1/\cos A + \sin A/\cos A)(1 - \sin A) = (1 + \sin A)/\cos A \times (1 - \sin A) = (1 - \sin^2 A)/\cos A = \cos^2 A/\cos A = \cos A$. Therefore $k = 1$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\cot \theta - \operatorname{cosec} \theta)^2 = (1 - \cos \theta)/(1 + \cos \theta)$

ANSWER: $(\cot \theta - \operatorname{cosec} \theta)^2 = (\cos \theta/\sin \theta - 1/\sin \theta)^2 = (\cos \theta - 1)^2/\sin^2 \theta = (1 - \cos \theta)^2/(1 - \cos^2 \theta) = (1 - \cos \theta)/(1 + \cos \theta)$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\sqrt{(\sec A - 1)/(\sec A + 1)} + \sqrt{(\sec A + 1)/(\sec A - 1)} = 2 \operatorname{cosec} A$

ANSWER: $= (\sec A - 1 + \sec A + 1)/\sqrt{(\sec^2 A - 1)} = 2\sec A/\tan A = 2/(\sin A) = 2\operatorname{cosec} A$.

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(1 + \cot^2 A)/(1 + \tan^2 A) = ((1 - \cot A)/(1 - \tan A))^2$

ANSWER: LHS = $(1 + \cos^2 A/\sin^2 A)/(1 + \sin^2 A/\cos^2 A) = ((\sin^2 A + \cos^2 A)/\sin^2 A)/((\cos^2 A + \sin^2 A)/\cos^2 A) = \cos^2 A/\sin^2 A = \cot^2 A$. RHS = $((1 - \cos A/\sin A)/(1 - \sin A/\cos A))^2 = ((\sin A - \cos A)/\sin A \times \cos A/(\cos A - \sin A))^2 = \cot^2 A$. Hence proved.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\sqrt{(\operatorname{cosec} A - 1)/(\operatorname{cosec} A + 1)} = \sec A - \tan A$

ANSWER: LHS = $\sqrt{((\operatorname{cosec} A - 1)/(\operatorname{cosec} A + 1) \times (\operatorname{cosec} A - 1)/(\operatorname{cosec} A - 1))} = \sqrt{((\operatorname{cosec} A - 1)^2/(\cot^2 A))} = (\operatorname{cosec} A - 1)/\cot A = 1/\sin A \times \tan A - 1/\cot A = \sec A - \tan A$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$

ANSWER: LHS = $\sin^2 \theta + 2\sin \theta \operatorname{cosec} \theta + \operatorname{cosec}^2 \theta + \cos^2 \theta + 2\cos \theta \sec \theta + \sec^2 \theta = (\sin^2 \theta + \cos^2 \theta) + 2(1+1) + (\operatorname{cosec}^2 \theta + \sec^2 \theta) = 1 + 4 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) = 7 + \tan^2 \theta + \cot^2 \theta$.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\cos A = 5/13$, verify that $\cos A/(1 - \tan A) + \sin A/(1 - \cot A) = \cos A + \sin A$.

ANSWER: $\sin A = 12/13$. LHS = $(5/13)/(1 - 5/12) + (12/13)/(1 - 12/5) = (5/13)/(7/12) + (12/13)/(-7/5) = 60/91 - 60/91 = 0$. RHS = $(5+12)/13 = 17/13$. Actually the identity is not correct as given. Wait - let me recalculate: $\cos A/(1 - \tan A) + \sin A/(1 - \cot A) = \cos A \times \cos A/(\cos A - \sin A) + \sin A \times \sin A/(\sin A - \cos A) = (\cos^2 A - \sin^2 A)/(\cos A - \sin A) = \cos A + \sin A$. Verified.

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CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\cos A - \cot A)^2 = (1 - \cos A)/(1 + \cos A)$

ANSWER: $(\cos A - \cot A)^2 = (\cos A - \cos A/\sin A)^2 = \cos^2 A(1 - 1/\sin A)^2$. This doesn't match.

Actually $\cot A = \cos A/\sin A$, so $\cos A - \cot A = \cos A(1 - 1/\sin A)$. The identity is: $(\cot A - \operatorname{cosec} A)^2 = (1 - \cos A)/(1 + \cos A)$. $(\cos A - \cot A)^2 = \cos^2 A(1 - 1/\sin A)^2 = \cos^2 A(\sin A - 1)^2/\sin^2 A = \cos^2 A(1 - \sin A)^2/\sin^2 A = (1 - \sin^2 A)(1 - \sin A)^2/\sin^2 A = (1 - \sin A)^3(1 + \sin A)/\sin^2 A$. Not equal to given. The correct identity is $(\cot A - \operatorname{cosec} A)^2 = (1 - \cos A)/(1 + \cos A)$.

Chapter 10: Heights and Distances

YEAR: 2024

CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A lamp post 9 m high casts a shadow $3\sqrt{3}$ m long on the ground. The Sun's elevation at this moment is:

OPTIONS: (A) 60° (B) 90° (C) 45° (D) 30°

ANSWER: (A) 60°

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A pole 6 m high casts a shadow $2\sqrt{3}$ m long on the ground, then the sun's elevation is:

OPTIONS: (A) 30° (B) 45° (C) 60° (D) 90°

ANSWER: (C) 60°

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A pole $7\sqrt{3}$ m high casts a shadow 21 m long on the ground, then the sun's elevation is:

OPTIONS: (A) 30° (B) 45° (C) 60° (D) 90°

ANSWER: (A) 30°

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A lamp post 15 m high casts a shadow $5\sqrt{3}$ m long on the ground. The Sun's elevation at this moment is:

OPTIONS: (A) 90° (B) 60° (C) 45° (D) 30°

ANSWER: (B) 60°

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it was 60° . Find the height of the tower and the length of original shadow.

ANSWER: Let height = h, original shadow = x. When altitude is 60° : $\tan 60^\circ = h/x \Rightarrow h = x\sqrt{3}$.
When altitude is 30° : $\tan 30^\circ = h/(x+40) \Rightarrow h = (x+40)/\sqrt{3}$. Equating: $x\sqrt{3} = (x+40)/\sqrt{3} \Rightarrow 3x = x+40 \Rightarrow 2x = 40 \Rightarrow x = 20$ m. $h = 20\sqrt{3} = 34.6$ m.

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The angles of depression of the top and the bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45° respectively. Find the height of the multi-storeyed building and the distance between the two buildings.

ANSWER: Let height of multi-storeyed building = h, distance between buildings = x. From top of multi-storeyed building to top of smaller building: $\tan 30^\circ = (h-8)/x \Rightarrow x = (h-8)\sqrt{3}$. From top to bottom: $\tan 45^\circ = h/x \Rightarrow x = h$. Equating: $h = (h-8)\sqrt{3} \Rightarrow h = \sqrt{3}h - 8\sqrt{3} \Rightarrow h(\sqrt{3}-1) = 8\sqrt{3} \Rightarrow h = 8\sqrt{3}/(\sqrt{3}-1) = 8\sqrt{3}(\sqrt{3}+1)/2 = 4(3+\sqrt{3}) = 18.93$ m. $x = h = 18.93$ m.

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: From a point on the ground, the angle of elevation of the top of a pedestal is 30° and that of the top of the flagstaff fixed on the pedestal is 60° . If the length of the flagstaff is 5 m, find the height of the pedestal and its distance from the point of observation on ground.

ANSWER: Let height of pedestal = h , distance = x . From ground to pedestal top: $\tan 30^\circ = h/x \Rightarrow x = h/\sqrt{3}$. To flagstaff top: $\tan 60^\circ = (h+5)/x \Rightarrow x = (h+5)/\sqrt{3}$. Equating: $h/\sqrt{3} = (h+5)/\sqrt{3} \Rightarrow 3h = h+5 \Rightarrow 2h = 5 \Rightarrow h = 2.5$ m. $x = 2.5\sqrt{3} = 4.33$ m.

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The angle of elevation of the top of a tower, 300 m high, from a point on the ground is observed as 30° . At an instant a hot air balloon passes vertically above the tower and at that instant its angle of elevation from same point is 60° . Find height of the balloon from the ground and distance of tower from point of observation.

ANSWER: Let distance = x . $\tan 30^\circ = 300/x \Rightarrow x = 300/\sqrt{3} = 519$ m. For balloon: $\tan 60^\circ = h/x \Rightarrow h = x\sqrt{3} = 300\sqrt{3} \times \sqrt{3} = 900$ m. Height of balloon = 900 m.

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: From a point on the ground, the angle of elevation of the top of a tree observed by a person is 60° . When moved back by 28 m, in the same line, the angle of elevation from another point becomes 30° . Find the height of the tree and its distance from the initial point.

ANSWER: Let height = h , initial distance = x . $\tan 60^\circ = h/x \Rightarrow h = x\sqrt{3}$. After moving back: $\tan 30^\circ = h/(x+28) \Rightarrow h = (x+28)/\sqrt{3}$. Equating: $x\sqrt{3} = (x+28)/\sqrt{3} \Rightarrow 3x = x+28 \Rightarrow 2x = 28 \Rightarrow x = 14$ m. $h = 14\sqrt{3} = 24.24$ m.

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A statue 3 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation

of the top of the pedestal is 30° . Find the height of the pedestal and its distance from the point of observation.

ANSWER: Let height of pedestal = h , distance = x . $\tan 30^\circ = h/x \Rightarrow x = h\sqrt{3}$. $\tan 60^\circ = (h+3)/x \Rightarrow x = (h+3)/\sqrt{3}$. Equating: $h\sqrt{3} = (h+3)/\sqrt{3} \Rightarrow 3h = h+3 \Rightarrow 2h = 3 \Rightarrow h = 1.5$ m. $x = 1.5\sqrt{3} = 2.6$ m.

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CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A drone is flying at a height of h metres. At an instant it observes the angle of elevation of top of an industrial turbine as 60° and angle of depression of foot of the turbine as 30° . If height of turbine is 200 metres, find h and distance of drone from the turbine.

ANSWER: Let distance = x . From drone to foot: $\tan 30^\circ = h/x \Rightarrow x = h\sqrt{3}$. From drone to top: $\tan 60^\circ = (200-h)/x \Rightarrow x = (200-h)/\sqrt{3}$. Equating: $h\sqrt{3} = (200-h)/\sqrt{3} \Rightarrow 3h = 200-h \Rightarrow 4h = 200 \Rightarrow h = 50$ m. $x = 50\sqrt{3} = 86.5$ m.

YEAR: 2024

CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: An injured bird was found on the roof of a building 15 m high. A fireman used an adjustable ladder making an angle of 60° with the ground to reach the roof. (i) Find the length of the ladder. (ii) Find the distance of the point on the ground where the ladder was fixed from the building. (iii)(a) To avoid skidding, the ladder was placed so the bottom touches the base of the wall opposite to the building, making an angle of 30° with the ground. Find the width of the road between the building and the wall. OR (b) Find the length of the ladder used in this case.

ANSWER: (i) $\sin 60^\circ = 15/l \Rightarrow l = 15/\sin 60^\circ = 15/(\sqrt{3}/2) = 30/\sqrt{3} = 10\sqrt{3}$ m. (ii) $\tan 60^\circ = 15/x \Rightarrow x = 15/\sqrt{3} = 5\sqrt{3}$ m. (iii)(a) $\tan 30^\circ = 15/y \Rightarrow y = 15\sqrt{3}$ m. (b) $\sin 30^\circ = 15/l \Rightarrow l = 30$ m.

Chapter 11: Areas Related to Circles

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The perimeter of the shaded region in the given figure is:

OPTIONS: (A) l (B) $l + a$ (C) $l + 2r$ (D) $l + 2r + a$

ANSWER: (C) $l + 2r$

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CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The ratio of the area of a quadrant of a circle to the area of the same circle is:

OPTIONS: (A) 1:2 (B) 2:1 (C) 1:4 (D) 4:1

ANSWER: (C) 1:4

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CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The length of arc subtending an angle of 210° at the centre of the circle, is $\frac{44}{3}$ cm. The radius of the circle is:

OPTIONS: (A) $2\sqrt{2}$ cm (B) 4 cm (C) 8 cm (D) $\frac{1}{4}$ cm

ANSWER: (B) 4 cm

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CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An arc of length 22 cm subtends an angle of x° at the centre of the circle. If radius of circle is 36 cm, the value of x is:

OPTIONS: (A) 35 (B) 40 (C) 60 (D) 30

ANSWER: (A) 35

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CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The area of a sector of angle θ (in degrees) of a circle with radius r is:

OPTIONS: (A) $\frac{\theta}{180} \times 2\pi r$ (B) $\frac{\theta}{180} \times \pi r^2$ (C) $\frac{\theta}{360} \times 2\pi r$ (D) $\frac{\theta}{720} \times 2\pi r^2$

ANSWER: (B) $\frac{\theta}{180} \times \pi r^2$

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CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The area of a quadrant of a circle whose circumference is 22 cm, is:

OPTIONS: (A) $11/4 \text{ cm}^2$ (B) $11/2 \text{ cm}^2$ (C) $33/4 \text{ cm}^2$ (D) $77/8 \text{ cm}^2$

ANSWER: (D) $77/8 \text{ cm}^2$

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The area of a smaller circle is equal to the area of a sector of a larger circle with central angle 120° . The radii of the smaller and larger circles are r and R respectively. Find $r:R$.

ANSWER: $\pi r^2 = 120/360 \times \pi R^2 \Rightarrow \pi r^2 = 1/3 \times \pi R^2 \Rightarrow r^2/R^2 = 1/3 \Rightarrow r:R = 1:\sqrt{3}$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Two concentric circles are of radii 6 cm and 10 cm. Find the length of the chord of the larger circle which touches the smaller circle.

ANSWER: Let chord AB of larger circle touch smaller circle at C. $OC = 6 \text{ cm}$, $OB = 10 \text{ cm}$. In $\triangle OCB$, $BC^2 = OB^2 - OC^2 = 100 - 36 = 64 \Rightarrow BC = 8 \text{ cm}$. $AB = 2BC = 16 \text{ cm}$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A chord of a circle of diameter 20 cm subtends an angle of 60° at the centre. Find the area of the corresponding minor segment.

ANSWER: Radius = 10 cm. Area of sector = $60/360 \times \pi \times 100 = 50\pi/3 \text{ cm}^2$. Area of equilateral triangle = $\sqrt{3}/4 \times 100 = 25\sqrt{3} \text{ cm}^2$. Area of segment = $50\pi/3 - 25\sqrt{3} = 50(3.14)/3 - 25(1.73) = 52.33 - 43.25 = 9.08 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the corresponding (i) minor sector (ii) major sector.

ANSWER: (i) Area of minor sector = $\frac{90}{360} \times \pi \times 100 = 25\pi = 78.5 \text{ cm}^2$. (ii) Area of major sector = $\frac{270}{360} \times \pi \times 100 = 75\pi = 235.5 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A chord of a circle of radius 10 cm subtends an angle of 60° at the centre. Find the area of the shaded region.

ANSWER: Area of sector = $\frac{60}{360} \times 3.14 \times 100 = 52.33 \text{ cm}^2$. Area of triangle = $\frac{\sqrt{3}}{4} \times 100 = 43.25 \text{ cm}^2$. Shaded area = $52.33 - 43.25 = 9.08 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A chord of a circle of radius 14 cm subtends an angle of 90° at the centre. Find perimeter of shaded region.

ANSWER: $AB = 14\sqrt{2} = 19.74 \text{ cm}$. Arc AB = $\frac{90}{360} \times 2 \times \frac{22}{7} \times 14 = 22 \text{ cm}$. Perimeter = $19.74 + 22 = 41.74 \text{ cm}$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A chord of a circle of radius 14 cm subtends an angle of 90° at the centre. Find the area of the corresponding minor and major segments.

ANSWER: Area of sector = $\frac{90}{360} \times \frac{22}{7} \times 196 = 154 \text{ cm}^2$. Area of triangle = $\frac{1}{2} \times 14 \times 14 = 98 \text{ cm}^2$. Minor segment = $154 - 98 = 56 \text{ cm}^2$. Major segment = $\pi r^2 - 56 = 616 - 56 = 560 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, OPRQ is a quadrant of a circle with centre O and radius 10 cm. If OP = 10 cm and OS = 6 cm, find the area of the shaded region.

ANSWER: Area of quadrant = $\frac{1}{4} \times \pi \times 100 = 25\pi = 78.5 \text{ cm}^2$. Area of $\Delta POS = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$. Shaded area = $78.5 - 30 = 48.5 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 11 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, OABC is a quadrant of a circle with centre O and radius 8 cm. If OD = 6 cm, find the perimeter of the shaded region.

ANSWER: Arc ABC = $\frac{90}{360} \times 2 \times \frac{22}{7} \times 8 = \frac{88}{7} \text{ cm}$. AD = $\sqrt{(8^2 - 6^2)} = \sqrt{28} = 2\sqrt{7} \text{ cm}$. CD = $8 - 6 = 2 \text{ cm}$. Perimeter = $\frac{88}{7} + 2\sqrt{7} + 2 = 24.57 \text{ cm}$.

Chapter 12: Surface Areas and Volumes

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The radius of a sphere is $\frac{7}{2} \text{ cm}$. The volume of the sphere is:

OPTIONS: (A) $231/3 \text{ cu cm}$ (B) $539/12 \text{ cu cm}$ (C) $539/3 \text{ cu cm}$ (D) 154 cu cm

ANSWER: (C) $539/3 \text{ cu cm}$

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The height and radius of a right circular cone are 24 cm and 7 cm respectively. The slant height of the cone is:

OPTIONS: (A) 24 cm (B) 31 cm (C) 26 cm (D) 25 cm

ANSWER: (D) 25 cm

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For which of the following solids is the lateral/curved surface area and total surface area the same?

OPTIONS: (A) Cube (B) Cuboid (C) Hemisphere (D) Sphere

ANSWER: (D) Sphere

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The total surface area of a solid hemisphere of radius 7 cm is:

OPTIONS: (A) $98\pi \text{ cm}^2$ (B) $147\pi \text{ cm}^2$ (C) $196\pi \text{ cm}^2$ (D) $228\frac{2}{3}\pi \text{ cm}^2$

ANSWER: (B) $147\pi \text{ cm}^2$

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The curved surface area of a cone with base radius 7 cm, is 550 cm^2 . The slant height of the cone is:

OPTIONS: (A) 25 cm (B) 14 cm (C) 20 cm (D) 24 cm

ANSWER: (A) 25 cm

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The total surface area of a cube of side 20 cm is:

OPTIONS: (A) 240 cm² (B) 160 cm² (C) 2400 cm² (D) 1600 cm²

ANSWER: (C) 2400 cm²

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Outer surface area of a cylindrical juice glass with radius 7 cm and height 10 cm, is:

OPTIONS: (A) 440 sq m (B) 594 sq m (C) 748 sq m (D) 1540 sq m

ANSWER: (A) 440 sq m

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The largest possible cone is just fitted inside a hollow cube of edge 25 cm. The radius of the base of the cone is:

OPTIONS: (A) 5 cm (B) 12.5 cm (C) 25 cm (D) 10 cm

ANSWER: (B) 12.5 cm

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A toy is in the form of a cone surmounted on a hemisphere. The cone and hemisphere have the same radii. The height of the conical part is equal to the diameter of its base. If the radius of the conical part is 5 cm, find the volume of the toy.

ANSWER: Radius = 5 cm, height of cone = 10 cm. Volume = volume of hemisphere + volume of cone = $\frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = \frac{2}{3}\pi(125) + \frac{1}{3}\pi(25 \times 10) = 250\pi/3 + 250\pi/3 = 500\pi/3 = 523.81 \text{ cm}^3$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A cubical block is surmounted by a hemisphere of radius 3.5 cm. What is the smallest possible length of the edge of the cube so that the hemisphere can totally lie on the cube? Find the total surface area of the solid so formed.

ANSWER: Edge of cube = diameter of hemisphere = 7 cm. Total surface area = $6a^2 + 2\pi r^2 - \pi r^2$
 $= 6a^2 + \pi r^2 = 6(49) + 22/7 \times 3.5 \times 3.5 = 294 + 38.5 = 332.5 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A solid is in the form of a cylinder with hemispherical ends of same radii. The total height is 20 cm and diameter of the cylinder is 14 cm. Find the surface area of the solid.

ANSWER: Radius = 7 cm, height of cylinder = $20 - 14 = 6$ cm? Wait total height = 20, with hemispheres at both ends. If diameter = 14, radius = 7. Height of cylinder = $20 - 2(7) = 6$ cm.
Surface area = curved area of cylinder + 2(curved area of hemisphere) = $2\pi rh + 2(2\pi r^2) = 2\pi(7)(6) + 4\pi(49) = 84\pi + 196\pi = 280\pi = 880 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A juice glass is cylindrical in shape with hemispherical raised up portion at the bottom. The inner diameter is 10 cm and height is 14 cm. Find the capacity of the glass.

ANSWER: Radius = 5 cm, height of cylinder = $14 - 5 = 9$ cm. Capacity = volume of cylinder - volume of hemisphere = $\pi r^2 h - 2/3\pi r^3 = \pi(25)(9) - 2/3\pi(125) = 225\pi - 250\pi/3 = (675\pi - 250\pi)/3 = 425\pi/3 = 425(3.14)/3 = 444.83 \text{ cm}^3$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A vessel is in the form of a hollow hemisphere surmounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.

ANSWER: Radius = 7 cm, height of cylinder = $13 - 7 = 6$ cm. Inner surface area = curved area of cylinder + curved area of hemisphere = $2\pi rh + 2\pi r^2 = 2\pi(7)(6) + 2\pi(49) = 84\pi + 98\pi = 182\pi = 572 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Two cubes each of volume 125 cm^3 are joined end to end. Find the volume and the surface area of the resulting cuboid.

ANSWER: Side of each cube = 5 cm. Volume of cuboid = $125 + 125 = 250 \text{ cm}^3$. Dimensions = $5 \times 5 \times 10$. Surface area = $2(5 \times 5 + 5 \times 10 + 5 \times 10) = 2(25 + 50 + 50) = 250 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A solid is in the shape of a cone surmounted on a hemisphere with both their diameters being equal to 7 cm and the height of the cone is equal to its radius. Find the volume of the solid.

ANSWER: Radius = 3.5 cm, height of cone = 3.5 cm. Volume = volume of cone + volume of hemisphere = $\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = \frac{1}{3}\pi(12.25)(3.5) + \frac{2}{3}\pi(42.875) = 42.875\pi/3 + 85.75\pi/3 = 128.625\pi/3 = 134.86 \text{ cm}^3$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A toy is in the form of a cone of radius 7 cm mounted on a hemisphere of same radius. The total height is 31 cm. Find the surface area of the toy.

ANSWER: Radius = 7 cm, height of cone = $31 - 7 = 24$ cm. Slant height of cone = $\sqrt{(24^2 + 7^2)} = \sqrt{(576 + 49)} = \sqrt{625} = 25$ cm. Surface area = curved area of cone + curved area of hemisphere = $\pi r l + 2\pi r^2 = \pi(7)(25) + 2\pi(49) = 175\pi + 98\pi = 273\pi = 858 \text{ cm}^2$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A spherical glass vessel has a cylindrical neck which is 7 cm long and 2 cm in diameter. The diameter of the spherical part is 14 cm. Find the capacity of the entire glass vessel.

ANSWER: Radius of sphere = 7 cm, radius of cylinder = 1 cm. Capacity = volume of sphere + volume of cylinder = $\frac{4}{3}\pi(7^3) + \pi(1^2)(7) = \frac{4}{3}\pi(343) + 7\pi = (\frac{1372\pi}{3} + 7\pi) = (\frac{1372\pi + 21\pi}{3}) = \frac{1393\pi}{3} = 1459.33 \text{ cm}^3$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A spherical glass vessel has a cylindrical neck 7 cm long and 8 cm in diameter. The radius of spherical part is 10 cm. Find the volume of the vessel.

ANSWER: Radius of cylinder = 4 cm. Volume = volume of sphere + volume of cylinder = $\frac{4}{3}\pi(10^3) + \pi(4^2)(7) = \frac{4000\pi}{3} + 112\pi = (\frac{4000\pi + 336\pi}{3}) = \frac{4336\pi}{3} = 4542.48 \text{ cm}^3$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: From each end of a solid cylinder of height 20 cm and base radius 7 cm, a cone of base radius 2.1 cm and height 5 cm is scooped out. Find the volume of the remaining solid.

ANSWER: Volume of cylinder = $\pi(7^2)(20) = 3080 \text{ cm}^3$. Volume of one cone = $\frac{1}{3}\pi(2.1^2)(5) = \frac{1}{3}\pi(4.41)(5) = 7.35\pi = 23.1 \text{ cm}^3$. Two cones = 46.2 cm^3 . Remaining = $3080 - 46.2 = 3033.8 \text{ cm}^3$.

YEAR: 2024

CHAPTER: 12 - Surface Areas and Volumes

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: A hemispherical bowl is packed in a cuboidal box. The bowl just fits in the box. Inner radius of the bowl is 10 cm. Outer radius is 10.5 cm. (i) Find the dimensions of the cuboidal box. (ii) Find the total outer surface area of the box. (iii)(a) Find the difference between the capacity of the bowl and the volume of the box. OR (b) The inner surface of the bowl and the thickness is to be painted. Find the area to be painted.

ANSWER: (i) Diameter = 21 cm, so box dimensions = $21 \times 21 \times 10.5 \text{ cm}$. (ii) Surface area = $2(21 \times 21 + 21 \times 10.5 + 21 \times 10.5) = 2(441 + 220.5 + 220.5) = 2(882) = 1764 \text{ cm}^2$. (iii)(a) Capacity = $\frac{2}{3} \times 3.14 \times 10^3 = 2093.33 \text{ cm}^3$. Volume of box = $21 \times 21 \times 10.5 = 4630.5 \text{ cm}^3$. Difference = 2537.17

cm³. (b) Area to paint = $2 \times 3.14 \times 10^2 + 3.14 \times (10.5^2 - 10^2) = 628 + 3.14 \times 10.25 = 628 + 32.185 = 660.185 \text{ cm}^2$.

Chapter 13: Statistics

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If a certain variable x divides a statistical data arranged in order into two equal parts, then the value of x is called the:

OPTIONS: (A) mean (B) median (C) mode (D) range

ANSWER: (B) median

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mean and median of a statistical data are 21 and 23 respectively. The mode of the data is:

OPTIONS: (A) 27 (B) 22 (C) 17 (D) 23

ANSWER: (A) 27

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The class mark of the median class of the following data is:

Class Interval: 10-25, 25-40, 40-55, 55-70, 70-85, 85-100

Frequency: 2, 3, 7, 6, 6, 6

OPTIONS: (A) 40 (B) 55 (C) 47.5 (D) 62.5

ANSWER: (D) 62.5

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The lower limit of the modal class in the following distribution is:

Runs Scored: 3000-4000, 4000-5000, 5000-6000, 6000-7000

Number of Batsmen: 5, 10, 9, 8

OPTIONS: (A) 3000 (B) 4000 (C) 5000 (D) 6000

ANSWER: (B) 4000

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The lower limit of the modal class in the following distribution is:

Income: 400-424, 425-449, 450-474, 475-499, 500-524

Number of workers: 12, 14, 8, 6, 10

OPTIONS: (A) 425 (B) 449 (C) 424.5 (D) 425.5

ANSWER: (A) 425

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The median group in the following frequency distribution is:

Class: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60

Frequency: 5, 8, 20, 15, 7, 5

OPTIONS: (A) 10-20 (B) 20-30 (C) 30-40 (D) 40-50

ANSWER: (C) 30-40

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the formula of mode given by $\text{mode} = l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \times h$, f_1 denotes the:

OPTIONS: (A) frequency of the modal class (B) frequency of class preceding modal class (C)

frequency of class succeeding modal class (D) cumulative frequency of modal class

ANSWER: (A) frequency of the modal class

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For a distribution, if mean = median = a, then its mode is:

OPTIONS: (A) 3a (B) 2a (C) a (D) 0

ANSWER: (C) a

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mode and mean of a data are 7 and 8 respectively. Then, the median of the data is:

OPTIONS: (A) $\frac{23}{3}$ (B) $\frac{13}{3}$ (C) $\frac{1}{13}$ (D) 33

ANSWER: (A) $\frac{23}{3}$

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If for a distribution, $\sum f_i x_i = 132 + 5p$, $\sum f_i = 20$ and mean is 8.1, then the value of p is:

OPTIONS: (A) 3 (B) 6 (C) 4 (D) 5

ANSWER: (B) 6

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following frequency distribution gives the monthly consumption of electricity of 68 consumers. Find the monthly mean consumption.

Monthly Consumption: 50-100, 100-150, 150-200, 200-250, 250-300, 300-350, 350-400

Number of Consumers: 4, 5, 13, 20, 14, 8, 4

ANSWER: Mean = 230.15 units. (Using assumed mean method with $A = 225$, $h = 50$, $\sum f_i u_i = 7/68$, Mean = $225 + 7/68 \times 50 = 230.15$)

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A life insurance agent found the following data for the distribution of 100 policy holders on the basis of their ages. Find the median age.

Age: 15-20, 20-25, 25-30, 30-35, 35-40, 40-45, 45-50, 50-55, 55-60

Number: 2, 4, 18, 21, 33, 11, 3, 6, 2

ANSWER: $N = 100$, $N/2 = 50$. Median class = 35-40. Median = $35 + (50-45)/33 \times 5 = 35.76$ years.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following table shows the ages of patients admitted in a hospital during a year. Find the median.

Age: 5-15, 15-25, 25-35, 35-45, 45-55, 55-65

Number of Patients: 6, 11, 21, 23, 14, 5

ANSWER: $N = 80$, $N/2 = 40$. Median class = 35-45. Median = $35 + (40-38)/23 \times 10 = 35.87$ years.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The lengths of 40 leaves of a plant are measured. Find the median length.

Length: 100-120, 120-140, 140-160, 160-180, 180-200

Number: 8, 9, 12, 5, 6

ANSWER: $N = 40$, $N/2 = 20$. Median class = 140-160. Median = $140 + (20-17)/12 \times 20 = 140 + 5 = 145$ mm.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A class teacher has the following absentees record of 30 students. If the mean number of days absent is 12, find x and y.

Number of days: 0-4, 4-8, 8-12, 12-16, 16-20, 20-24

Number of students: 1, 8, x, 6, 5, y

ANSWER: Total students: $1+8+x+6+5+y = 30 \Rightarrow x+y = 10$. Mean = $(2 \times 1 + 6 \times 8 + 10x + 14 \times 6 + 18 \times 5 + 22y)/30 = 12 \Rightarrow 2+48+10x+84+90+22y = 360 \Rightarrow 224+10x+22y = 360 \Rightarrow 10x+22y = 136 \Rightarrow 5x+11y = 68$. Solving: $x=7, y=3$.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following data gives the information about the lifetimes of 225 neon lamps. Find the median lifetime.

Lifetime: 1500-2000, 2000-2500, 2500-3000, 3000-3500, 3500-4000, 4000-4500

Number of lamps: 10, 35, 52, 61, 38, 29

ANSWER: $N = 225, N/2 = 112.5$. Median class = 3000-3500. Median = $3000 + (112.5-97)/61 \times 500 = 3000 + 15.5/61 \times 500 = 3000 + 127.05 = 3127.05$ hours.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following distribution shows the weekly pocket allowance (in ₹) of some children. The mean pocket allowance is ₹180. Find the value of f. Hence find the mode.

Weekly Pocket Allowance: 110-130, 130-150, 150-170, 170-190, 190-210, 210-230, 230-250

Number of Children: 7, 6, 9, 13, f, 5, 4

ANSWER: $\Sigma f = 44+f, \Sigma fx = 7520 + 200f$. Mean = $(7520+200f)/(44+f) = 180 \Rightarrow 7520+200f = 7920+180f \Rightarrow 20f = 400 \Rightarrow f = 20$. Mode class = 190-210. Mode = $190 + (20-13)/(40-13-5) \times 20 = 190 + 7/22 \times 20 = 196.36$.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: Mutual Fund: Net Asset Value (NAV) per unit of mutual fund of ICICI mutual funds. Find (i) upper limit of modal class (ii) median class (iii)(a) mode NAV or (b) median NAV.

NAV: 0-5, 5-10, 10-15, 15-20, 20-25

Number of funds: 13, 16, 22, 18, 11

ANSWER: (i) Modal class = 10-15, upper limit = 15. (ii) $N = 80$, $N/2 = 40$. Median class = 10-15. (iii)(a) Mode = $10 + (22-16)/(44-16-18) \times 5 = 10 + 6/10 \times 5 = 13$. (b) Median = $10 + (40-29)/22 \times 5 = 10 + 11/22 \times 5 = 12.5$.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: Resident Welfare Association installed three electric poles A, B and C in the park. Find (i) position of pole C (ii) distance of pole B from corner O (iii)(a) position of fourth pole D to form a parallelogram ABCD or (b) distance between poles A and C.

ANSWER: (i) $C = (12, 7)$ (from graph). (ii) $OB = \sqrt{4^2 + 8^2} = \sqrt{80} = 4\sqrt{5}$. (iii)(a) For parallelogram ABCD, $D = A + C - B = (2, 2) + (12, 7) - (4, 8) = (10, 1)$. (b) $AC = \sqrt{(12-2)^2 + (7-2)^2} = \sqrt{100+25} = \sqrt{125} = 5\sqrt{5}$.

Chapter 14: Probability

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A card is drawn from a well shuffled deck of 52 playing cards. The probability that drawn card is a red queen, is:

OPTIONS: (A) $1/13$ (B) $2/13$ (C) $1/52$ (D) $1/26$

ANSWER: (D) $1/26$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two dice are rolled together. The probability of getting a doublet is:

OPTIONS: (A) $2/36$ (B) $1/36$ (C) $1/6$ (D) $5/6$

ANSWER: (C) $1/6$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If probability of winning a game is p , then probability of losing the game is:

OPTIONS: (A) $1+p$ (B) $-p$ (C) $p-1$ (D) $1-p$

ANSWER: (D) $1-p$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A die is thrown once. The probability of getting a number less than 6 is:

OPTIONS: (A) 0 (B) $\frac{5}{6}$ (C) $\frac{1}{6}$ (D) 1

ANSWER: (B) $\frac{5}{6}$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Three coins are tossed together. The probability of getting exactly one tail, is:

OPTIONS: (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{7}{8}$ (D) $\frac{3}{8}$

ANSWER: (D) $\frac{3}{8}$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The total number of outcomes in the experiment of simultaneous throw of three dice is:

OPTIONS: (A) 6 (B) 18 (C) 36 (D) 216

ANSWER: (D) 216

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $P(A)$ denotes the probability of an event A , then:

OPTIONS: (A) $P(A) < 0$ (B) $P(A) > 1$ (C) $0 \leq P(A) \leq 1$ (D) $-1 \leq P(A) \leq 1$

ANSWER: (C) $0 \leq P(A) \leq 1$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A pair of dice is thrown simultaneously. The number of outcomes in favour of event E that the sum of numbers on both dice is at least 9, is:

OPTIONS: (A) 4 (B) 6 (C) 10 (D) 26

ANSWER: (C) 10

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A black card is lost from a deck of 52 playing cards. Rest of the cards are shuffled and one card is drawn. The probability that drawn card is king of hearts, is:

OPTIONS: (A) $1/52$ (B) $1/4$ (C) $1/51$ (D) $1/26$

ANSWER: (C) $1/51$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two dice are rolled together. The probability that only one die shows number 4, is:

OPTIONS: (A) $11/36$ (B) $1/3$ (C) $5/18$ (D) $1/4$

ANSWER: (C) $5/18$

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If probability of happening of an event is 57%, then probability of non-happening of the event is:

OPTIONS: (A) 0.43 (B) 0.57 (C) 53% (D) 1/57

ANSWER: (A) 0.43

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A bag contains 40 marbles out of which some are white and others are black. If the probability of drawing a black marble is $\frac{3}{5}$, find the number of white marbles.

ANSWER: Let number of black marbles = n . $\frac{n}{40} = \frac{3}{5} \Rightarrow n = 24$. White marbles = $40 - 24 = 16$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In a pre-primary class, a teacher put cards numbered 20 to 59 in a bowl. A student picked up a card at random. Find the probability that the number read was (i) a prime number (ii) a perfect square.

ANSWER: Total cards = 40. (i) Prime numbers from 20 to 59: 23, 29, 31, 37, 41, 43, 47, 53, 59 = 9 numbers. Probability = $\frac{9}{40}$. (ii) Perfect squares: 25, 36, 49 = 3 numbers. Probability = $\frac{3}{40}$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A box contains 120 discs numbered from 1 to 120. If one disc is drawn at random, find the probability that (i) it bears a 2-digit number (ii) the number is a perfect square.

ANSWER: Total discs = 120. (i) 2-digit numbers from 10 to 99 = 90. Probability = $\frac{90}{120} = \frac{3}{4}$. (ii) Perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100 = 10. Probability = $\frac{10}{120} = \frac{1}{12}$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A coin is dropped at random on a rectangular region. What is the probability that it will land inside the circle with radius 0.7 m?

ANSWER: Area of rectangle = $3 \times 2 = 6 \text{ m}^2$. Area of circle = $\pi(0.7)^2 = \frac{22}{7} \times 0.49 = 1.54 \text{ m}^2$.
Probability = $1.54/6 = 77/300$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A die is thrown twice. What is the probability that (i) difference between two numbers obtained is 3 (ii) sum of the numbers obtained is 8?

ANSWER: Total outcomes = 36. (i) Pairs with difference 3: (1,4),(4,1),(2,5),(5,2),(3,6),(6,3) = 6. Probability = $6/36 = 1/6$. (ii) Pairs with sum 8: (2,6),(3,5),(4,4),(5,3),(6,2) = 5. Probability = $5/36$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: 15 defective pens are accidentally mixed with 145 good ones. One pen is taken out at random. Determine the probability that the pen taken out is a good one.

ANSWER: Total pens = 160. Good pens = 145. Probability = $145/160 = 29/32$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: There are 80 cards numbered from 1 to 80. One card is drawn at random. Find the probability that the number on the selected card is not divisible by 8.

ANSWER: Numbers divisible by 8 up to 80: 8,16,24,32,40,48,56,64,72,80 = 10 numbers. Not divisible by 8 = 70 numbers. Probability = $70/80 = 7/8$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A lot consists of 200 pens of which 180 are good and the rest are defective. A customer will buy a pen if it is not defective. The shopkeeper draws a pen at random and gives it to the customer. What is the probability that the customer will not buy it? Another lot of 100 pens containing 80 good pens is mixed with the previous lot. The shopkeeper now draws one pen at random from the entire lot. What is the probability that the customer will buy the pen?

ANSWER: First lot: defective = 20. $P(\text{not buy}) = 20/200 = 1/10$. After mixing: total = 300, good = $180+80 = 260$. $P(\text{buy}) = 260/300 = 13/15$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A bag contains 6 blue, 4 white and 8 red marbles. A marble is drawn at random. Find the probability that the marble so drawn is (i) white (ii) white or red (iii) not red.

ANSWER: Total = 18. (i) $P(\text{white}) = 4/18 = 2/9$. (ii) $P(\text{white or red}) = (4+8)/18 = 12/18 = 2/3$. (iii) $P(\text{not red}) = (6+4)/18 = 10/18 = 5/9$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A game of chance consists of spinning a wheel which comes to rest at one of the numbers from 1 to 10 with equal probabilities. What is the probability that the wheel stops at (i) a prime number greater than 2 (ii) an odd number less than 9 (iii) a multiple of 4?

ANSWER: Total outcomes = 10. (i) Prime numbers > 2 : 3, 5, 7 = 3. $P = 3/10$. (ii) Odd numbers < 9 : 1, 3, 5, 7 = 4. $P = 4/10 = 2/5$. (iii) Multiples of 4: 4, 8 = 2. $P = 2/10 = 1/5$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A card is drawn at random from a well-shuffled deck of 52 playing cards. Find the probability that the card drawn is (i) a red king (ii) not a black card (iii) an ace of hearts.

ANSWER: Total = 52. (i) Red kings: 2 (hearts king, diamonds king). $P = 2/52 = 1/26$. (ii) Not black = red cards = 26. $P = 26/52 = 1/2$. (iii) Ace of hearts: 1. $P = 1/52$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: From a well-shuffled deck of 52 playing cards, all black queens and red kings are removed. One card is selected at random from the remaining cards. Find the probability that the selected card is (i) an ace (ii) a jack of red colour (iii) a king of spade.

ANSWER: Removed: 2 black queens + 2 red kings = 4 cards. Remaining = 48. (i) Aces: 4. $P = \frac{4}{48} = \frac{1}{12}$. (ii) Jack of red colour: 2. $P = \frac{2}{48} = \frac{1}{24}$. (iii) King of spade: 1. $P = \frac{1}{48}$.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: E

ORIGINAL_MARKS: 4

QUESTION: Family structure: 51% families had no children, 20% had one child, 19% had two children, 7% had three children and 3% had four or more children. A family is selected at random. (i) Find the probability that the selected family has two or three children. (ii) Find the probability that the selected family has more than one child. (iii)(a) Find the probability that the selected family has less than three children. OR (b) Find the probability that the selected family has more than two children.

ANSWER: (i) $P(2 \text{ or } 3) = 19\% + 7\% = 26\% = 0.26$. (ii) $P(\text{more than one}) = 19\% + 7\% + 3\% = 29\% = 0.29$. (iii)(a) $P(\text{less than three}) = 51\% + 20\% + 19\% = 90\% = 0.90$. (b) $P(\text{more than two}) = 7\% + 3\% = 10\% = 0.10$.

Assertion-Reason Questions (Chapter-wise distribution)

YEAR: 2024

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The pair of linear equations $5x + 2y + 6 = 0$ and $7x + 6y + 18 = 0$ have infinitely many solutions. Reason (R): The pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ have infinitely many solutions, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (D) Assertion (A) is false, but Reason (R) is true.

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The probability of getting number 8 on rolling a die is zero (0).

Reason (R): The probability of an impossible event is zero (0).

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): If PA and PB are tangents drawn to a circle with centre O from an external point P, then quadrilateral OAPB is a cyclic quadrilateral. Reason (R): In a cyclic quadrilateral, opposite angles are equal.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2024

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Zeroes of polynomial $p(x) = x^2 - 2x - 3$ are -1 and 3. Reason (R): The graph of polynomial $p(x) = x^2 - 2x - 3$ intersects x-axis at (-1, 0) and (3, 0).

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

YEAR: 2024

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): For any two natural numbers a and b , the HCF of a and b is a factor of the LCM of a and b . Reason (R): HCF of any two natural numbers divides both the numbers.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): All congruent triangles are similar. Reason (R): In congruent triangles, the ratio of corresponding sides is 1:1.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

YEAR: 2024

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The prime numbers which divide 36 also divide 6. Reason (R): Any number which divides p^2 also divides p .

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is

not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.
ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): For $\sin \theta = 1$, $\cos \theta$ must be 0. Reason (R): $\sin^2 \theta - \cos^2 \theta = 1$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2024

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Every quadratic equation has two real roots. Reason (R): A quadratic polynomial can have at most two zeroes.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (D) Assertion (A) is false, but Reason (R) is true.

YEAR: 2024

CHAPTER: 13 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): Median marks of students in a class test is 16. It means half of the class got marks less than 16. Reason (R): Median divides the distribution in two equal parts.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

YEAR: 2024

CHAPTER: 14 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): If E is an event such that $P(E) = 1/999$, then $P(\bar{E}) = 0.001$. Reason (R): $P(E) + P(\bar{E}) = 1$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (D) Assertion (A) is false, but Reason (R) is true.

YEAR: 2024

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): When a hemisphere of same radius (r) is carved out from one side of a solid wooden cylinder, the total surface area of remaining solid is increased by $2\pi r^2$. Reason (R): Curved surface area of hemisphere is $2\pi r^2$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (D) Assertion (A) is false, but Reason (R) is true.

YEAR: 2024

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): $(a + \sqrt{b}) \cdot (a - \sqrt{b})$ is a rational number, where a and b are positive integers. Reason (R): Product of two irrationals is always rational.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): $\triangle ABC \sim \triangle PQR$ such that $\angle A = 65^\circ$, $\angle C = 60^\circ$. Hence $\angle Q = 55^\circ$.

Reason (R): Sum of all angles of a triangle is 180° .

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The point (0, 4) lies on y-axis. Reason (R): The x-coordinate of a point lying on y-axis is zero.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

YEAR: 2024

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): A line drawn parallel to any one side of a triangle intersects the other two sides in the same ratio. Reason (R): Parallel lines cannot be drawn to any side of a triangle.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2024

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The distance of $P(a,b)$ from origin is $a^2 + b^2$. Reason (R): The distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (D) Assertion (A) is false, but Reason (R) is true.

YEAR: 2024

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): $\sqrt{2}(5 - \sqrt{2})$ is an irrational number. Reason (R): Product of two irrational numbers is always irrational.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): For an acute angle θ , $\cot \theta = 1 \Rightarrow \cos \theta = \sqrt{2}$. Reason (R): $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (D) Assertion (A) is false, but Reason (R) is true.

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): For an angle θ , $\sec \theta = 1 \Rightarrow \tan \theta = 0$. Reason (R): $\sec^2 \theta + \tan^2 \theta = 1$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2024

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): In a right angle triangle ABC, $\angle B = 90^\circ$. Therefore the value of $\cos(A+C)$ is equal to 0. Reason (R): $A + B + C = 180^\circ$ and $\cos 90^\circ = 0$.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).